Entropy contraction on complete bipartite graphs via reverse erasure: only the rarest symbol matters

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Abstract

Caputo, Chen, Gu and Polyanskiy conjectured that one step of the half-lazy random walk on the complete bipartite graph $K_{n,n}$ shrinks relative entropy by at most the factor $\log n/[2\log(2n)]$, with single vertices as the only worst starting points. They proved the case $n = 3$ with computer assistance. We prove the conjecture for every n, and for $K_{m,n}$ with $M = \max(m, n)$ in place of n. The key observation is that one step of this walk splits into two copies of a simple channel, the Bayesian reverse of an erasure channel: keep the current symbol with probability α, otherwise replace it by a fresh draw from a prior ρ. For this channel we show that the relative-entropy contraction coefficient depends on the prior only through its smallest atom q, at every retention probability. Up to the sharp threshold $\alpha = q^{q/(1-q)}$ the coefficient is $\log(1/q)/\log(1/(\alpha q))$, which is a translation of a mean inequality of Gao; above the threshold the worst inputs become full-support exponential tilts. Further consequences are a rank-one weighted bipartite extension (which also settles a numerical observation on the lazy star), an extension to block-reset chains, and the exact one-way communication rate for agreement distillation when the party holding the erased copy speaks.

Keywords. Strong data processing; relative entropy; contraction coefficient; reverse erasure; Markov chains; complete bipartite graph; weighted means; agreement distillation.

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1 Introduction

1.1 Why entropy contraction coefficients?

Let K be a Markov kernel on a finite set and π a probability law with full support. If a random state with law v is moved one step by K, its relative entropy from πK can only decrease. The *contraction coefficient* records the worst-case fraction that survives:

$$\eta_{\mathrm{KL}}(\pi, K) = \sup_{v \neq \pi} \frac{D(vK\|\pi K)}{D(v\|\pi)}, \qquad D(v\|\pi) = \sum_x v_x \log \frac{v_x}{\pi_x}. \tag{1.1}$$

All logarithms are natural unless stated otherwise, and $0 \log 0 = 0$. The reference law π is fixed throughout; maximising also over π is a different problem [15]. An *extremiser* is a law $v \neq \pi$ at which the supremum in (1.1) is attained.

Inequalities of the form $D(vK\|\pi K) \leq c\, D(v\|\pi)$ are *strong data-processing inequalities*. They go back to Ahlswede and Gács [1], and are surveyed in [16, 17]. In Markov-chain language, $1 - \eta_{\mathrm{KL}}(\pi, P)$ is a one-step rate of entropy decay towards the stationary law π. A useful benchmark is the *local* (χ^2) coefficient, obtained by letting v approach π; for a reversible chain it is the square of the second-largest absolute eigenvalue. The global coefficient (1.1) is never smaller, and can be strictly larger; comparisons between such coefficients are developed in [13, 17].

1.2 The conjecture, and why it was posed

Caputo, Chen, Gu and Polyanskiy (CCGP) [4] compare three ways of measuring entropy decay: half-step, full-step and continuous-time. They point out that, unlike the extremal functions for log-Sobolev-type constants, extremal distributions for η_{KL} can fail to have full support. Their illustration for the full-step coefficient is the half-lazy walk on $K_{n,n}$: stay put with probability $1/2$, otherwise jump to a uniformly chosen vertex on the other side. (This is their lazy random walk, Definition 2.4 in the authors' manuscript and Definition 4 in arXiv v1.) In Example 4.4 (arXiv v1: Example 28) they report that numerical computation suggests

$$\eta_{\mathrm{KL}}(\pi, P) = \frac{\log n}{2\log(2n)}, \qquad n \geq 3, \tag{1.2}$$

with equality only at point distributions. Their Proposition 4.5 (arXiv v1: Proposition 29) proves $n = 3$ by a computer-assisted case analysis. The text following the proof of Proposition 4.5 conjectures the statement for all $n \geq 3$ and says that it is unclear how to prove it in general.

The point of the example is therefore *which inputs are worst*, not how fast the walk mixes. The walk on $K_{n,n}$ has spectral gap $1/2$; its one-step entropy decay rate $1 - \eta_{\mathrm{KL}}$ lies between $3/4$ ($n = 2$) and $1/2$ ($n \to \infty$) for $n \geq 2$. Our results give no new mixing-time information for this chain. What they do give is an exact constant, the exact set of worst starting states (single vertices, the "cold start"), and an explicit family in which the global coefficient tends to $1/2$ while the local coefficient stays at $1/4$.

1.3 The idea in one paragraph

Watch the walk from one side of the graph. With probability $1/2$ a walker on side A stays where it is; otherwise it leaves, and the walkers arriving on side A from side B land at a fresh uniform vertex of A. So the conditional law on side A after one step is "keep the old symbol with some probability, otherwise draw a fresh one from the prior". That is the Bayesian reverse of an erasure channel. An erasure channel keeps a symbol with probability α and otherwise outputs the blank symbol e; its reverse sends a kept symbol to itself and fills a blank with a fresh sample. Section 6 makes this exact: one step of the half-lazy bipartite walk is a pair of reverse-erasure channels at retention $\alpha = 1/2$, with one bookkeeping term that is paid only once. The whole problem is thus reduced to computing the contraction coefficient of reverse erasure, which Sections 2–5 do for every prior and every retention probability.

1.4 What is new and what is inherited

Given Gao's weighted mean inequality [7, 8], the conjecture (1.2) itself needs only two ingredients: the uniform-prior reverse-erasure coefficient at $\alpha = 1/2$ (Theorem 2.5 with uniform ρ, whose upper bound *is* Gao's inequality) and the two-part decomposition of Section 6. We say this plainly because the analytic core of the closed formula is not ours. The new content is:

(a) the reduction to the smallest atom for *every* prior and *every* retention probability, including the regime with no closed formula (Theorem 2.3, Proposition 2.4);

(b) the sharpness of the threshold, the complete equality analysis, and the description of extremisers above it (Theorems 2.5 and 2.7);

(c) the two-part decomposition, which proves the conjecture, extends it to unequal parts and rank-one cross weights, and settles the lazy star $K_{1,N}$, left as a numerical observation by Jadhav [12] (Theorem 6.1, Corollary 6.2);

(d) Section 7, which extends the reduction to block-reset chains, including arbitrary laziness, matrix powers, stars and multipartite graphs;

(e) the exact one-way agreement-distillation rate for a general prior (Corollary 8.1), a general-prior form of the uniform binary case evaluated by Zhao [19].

The reverse binary-erasure channel and its usefulness as an SDPI example appear in Weinberger and Shayevitz [18]. The complete uniform binary coefficient is known: the high-retention branch follows from Nair and Wang [14] and the low-retention branch is Zhao's [19, Theorem 3.5.4]; Appendix A recalls both. Our bounded literature search (Section 10) found no earlier statement of (a), of an all-n proof of (1.2), or of a non-uniform closed form for this channel. Priority remains subject to specialist review.

1.5 Results at a glance

Table 1 lists the results in the order in which they are proved. Here q is the smallest prior atom, α the retention probability, and $\alpha_*(q) = q^{q/(1-q)}$ the threshold.

Result	In words	New or known	Proof
Theorem 2.3, Proposition 2.4	The reverse-erasure coefficient depends on the prior only through q; an exact one-variable test decides it at every α.	New	§3
Theorem 2.5	For $\alpha \le \alpha_*(q)$ the coefficient is $\log(1/q)/\log(1/(\alpha q))$, attained only at a rarest point input; the threshold is sharp.	Upper bound is Gao's inequality; sharpness and general-prior form new	§4
Proposition 2.6, Theorem 2.7	Local coefficient α; above the threshold the worst inputs are full-support exponential tilts.	New	§5
Theorem 6.1, Corollary 6.2	Half-lazy $K_{m,n}$: $\log M/[2\log(2M)]$, only point extremisers for $M \ge 3$; rank-one weights; lazy star.	Conjecture of [4]; proof new	§6
Section 7	Coefficient of any block-reset chain equals that of a chain with at most two states per block (laziness, powers, stars, multipartite graphs); values there are computed numerically.	New	§7
Corollary 8.1	Below the threshold, the exact one-way rate for agreement distillation is $\log(1/\alpha)/\log(1/(\alpha q))$.	Uniform binary case known [19]; general prior new	§8

Table 1: Main results. "New" means that no earlier statement was found in the bounded search described in Section 10.

Worked example: $K_{3,3}$. The stationary law is uniform, $\pi_x = 1/6$. Start at a single vertex x. Then $D(\delta_x\|\pi) = \log 6$. After one step the walker is at x with probability $1/2$ and at each of the three opposite vertices with probability $1/6$, so

$$D(\delta_x P\|\pi) = \tfrac{1}{2}\log\frac{1/2}{1/6} + 3\cdot\tfrac{1}{6}\log 1 = \tfrac{1}{2}\log 3, \qquad \frac{D(\delta_x P\|\pi)}{D(\delta_x\|\pi)} = \frac{\log 3}{2\log 6} \approx 0.3066.$$

This is the value in (1.2) and in CCGP's Proposition 4.5. In the reverse-erasure picture, the conditional prior on each side is uniform with $q = 1/3$, the retention probability is $\alpha = 1/2$, and $1/2 \le \alpha_*(1/3) = 3^{-1/2} \approx 0.577$, so the closed formula applies (Figure 1, right). The local coefficient is only $1/4$.

Roadmap. Section 2 sets up the channel and states the channel results. Section 3 proves the reduction to the smallest atom, Section 4 the closed formula and its sharp threshold, and Section 5

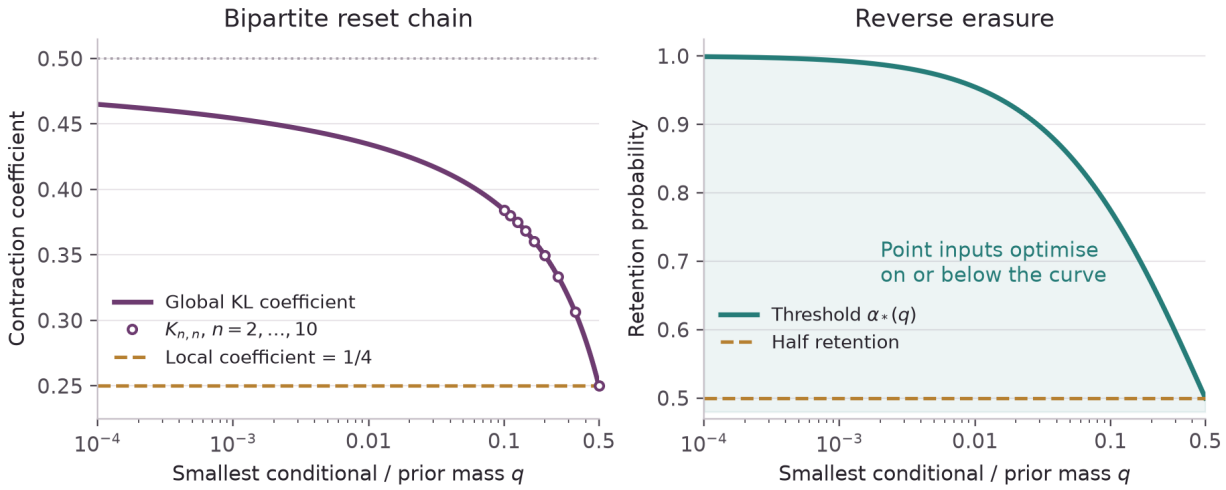


Figure 1: Exact formulas, not numerical fits. Left: the global coefficient $\log(1/q)/[2\log(2/q)]$ of a half-lazy bipartite reset chain rises towards $1/2$ as the smallest conditional mass q tends to zero; the circles mark $K_{n,n}$, $q = 1/n$, $n = 2, \ldots, 10$. The local quadratic coefficient stays at $1/4$. Right: the reverse-erasure threshold $\alpha_*(q) = q^{q/(1-q)}$. Retention $1/2$ always lies on or below it, which is why the bipartite walk is always in the regime with a closed formula.

the regime above the threshold. Section 6 proves the bipartite theorem and Section 7 extends it to block-reset chains. Section 8 gives the communication application. Section 9 lists open problems and Section 10 records how the results were checked. Appendices give the known uniform binary case, standard tensorisation and the numerical diagnostics.

1.6 Notation

Symbol	Meaning
$\rho = (\rho_1, \ldots, \rho_n)$	prior on the ordinary symbols, all $\rho_i > 0$, $n \geq 2$
$q = \min_i \rho_i \in (0, 1/2]$	smallest atom ("rarest symbol")
$\alpha \in (0, 1)$	retention probability of the erasure channel
R_ρ	reverse-erasure kernel (2.1)
$m_{\rho,\alpha} = (\alpha\rho, 1-\alpha)$	input reference law, with $m_{\rho,\alpha} R_\rho = \rho$
$\eta(\rho, \alpha)$	$\eta_{\mathrm{KL}}(m_{\rho,\alpha}, R_\rho)$, the coefficient studied here
$c_0(q, \alpha)$	$\log(1/q)/\log(1/(\alpha q))$, the point-input ratio
$\alpha_*(q)$	$q^{q/(1-q)}$, the point-optimality threshold
$p = 1/c$	dual exponent attached to a candidate coefficient c
$G_\rho(z)$	weighted geometric mean $\prod_i z_i^{\rho_i}$
$H_{w,\alpha,p}(r)$	one-variable dual test function (2.5)
$d(t\|\|s)$	binary relative entropy

2 The reverse-erasure channel and the channel results

2.1 The channel

Let $\rho = (\rho_1, \ldots, \rho_n)$ be positive, $n \geq 2$. The reverse-erasure kernel R_ρ, from $[n] \cup \{e\}$ to $[n]$, is

$$R_\rho(i, \cdot) = \delta_i, \qquad R_\rho(e, \cdot) = \rho. \tag{2.1}$$

For $0 < \alpha < 1$, its input reference is

$$m_{\rho,\alpha} = (\alpha\rho_1, \ldots, \alpha\rho_n, 1-\alpha), \qquad m_{\rho,\alpha} R_\rho = \rho. \tag{2.2}$$

If $X \sim \rho$ and E equals X with probability α and e otherwise, then $E \sim m_{\rho,\alpha}$ and R_ρ is the conditional law of X given E: the Bayesian reverse of the erasure channel. Set

$$\eta(\rho,\alpha) = \eta_{\mathrm{KL}}(m_{\rho,\alpha}, R_\rho), \qquad q = \min_i \rho_i \in (0, 1/2].$$

Writing any input as $v = (tu, 1-t)$, with u a law on $[n]$ and $t \in [0,1]$ its ordinary mass, gives

$$vR_\rho = tu + (1-t)\rho, \qquad D(v\|m_{\rho,\alpha}) = d(t\|\alpha) + tD(u\|\rho). \tag{2.3}$$

Remark 2.1 (Where the channel appears). (i) Composing the erasure channel with its reverse gives the lazy reset kernel $\alpha I + (1-\alpha)\mathbf{1}\rho^{\mathsf{T}}$: keep the symbol, or resample it from ρ. For uniform ρ this is the Potts channel studied by Gu and Polyanskiy [10], which therefore factors as erasure followed by reverse erasure. (ii) In sampling language, erasure is the "down" step and R_ρ the "up" (resampling) step of a down–up walk. For a product law $\rho^{\otimes N}$ with each coordinate erased independently, tensorisation (Appendix B) shows that $\eta(\rho,\alpha)$ is exactly the KL contraction coefficient of the up step of this block resampler, with the law of the erased configuration as reference. For non-product targets the up step resamples from conditional laws rather than from ρ, so our results give only the product-measure baseline. The proximal sampler [5] is a continuous analogue (noising forward, posterior denoising backward). (iii) The coefficient $\eta(\rho,\alpha)$ is the quantity s^* attached to the pair (erased observation, clean symbol) in the hypercontractivity language of Anantharam, Gohari, Kamath and Nair [2]; Zhao's evaluation [19, Theorem 3.5.4] is stated in that language.

2.2 The dual criterion and the test function H

A candidate coefficient $c \in (0,1)$ can be tested without searching over inputs. Put $p = 1/c$ and $G_\rho(z) = \prod_i z_i^{\rho_i}$. By entropy duality (Lemma 3.1, proved in Section 3), $\eta(\rho,\alpha) \le c$ holds if and only if

$$\alpha \sum_i \rho_i z_i^p + (1-\alpha) G_\rho(z)^p \le \left(\sum_i \rho_i z_i\right)^p \qquad \text{for all } z \in [0,\infty)^n. \tag{2.4}$$

This is an inequality between weighted means. It turns out that only test vectors with one large coordinate matter. Evaluating (2.4) at $z = (r, 1, \dots, 1)$, with the large coordinate at an atom of weight w, and dividing the left side by the right side, gives the one-variable function

$$H_{w,\alpha,p}(r) = \frac{\alpha(wr^p + 1 - w) + (1-\alpha)r^{pw}}{(wr + 1 - w)^p}, \qquad r \ge 1. \tag{2.5}$$

We have $H_{w,\alpha,p}(1) = 1$. As $r \to \infty$ the test vector, after normalisation, concentrates on one coordinate; this limit corresponds to a point input, and $H_{w,\alpha,p}(\infty) := \lim_{r\to\infty} H_{w,\alpha,p}(r) = \alpha w^{1-p}$.

Definition 2.2 (Dual extremiser)**.** Let $p_* = 1/\eta(\rho,\alpha)$. A *dual extremiser* is a nonconstant vector $z \in [0,\infty)^n \setminus \{0\}$ at which (2.4) holds with equality for $p = p_*$. It is *positive* if every coordinate is strictly positive.

2.3 Statements

Theorem 2.3 (Only the rarest symbol matters)**.** *For every full-support finite prior ρ and every $0 < \alpha < 1$,*

$$\eta(\rho,\alpha) = \eta((q, 1-q), \alpha), \qquad q = \min_i \rho_i. \tag{2.6}$$

In plain words. Erasing and refilling forgets the same fraction of information, in the worst case, for every prior whose rarest symbol has probability q. For example, the priors $(0.1, 0.2, 0.7)$ and $(0.1, 0.9)$ have exactly the same coefficient at every retention probability (although their channels and extremisers differ).

Proposition 2.4 (Exact scalar criterion)**.** *Let $0 < c < 1$ and $p = 1/c$. Then $\eta(\rho, \alpha) \le c$ if and only if*

$$H_{q,\alpha,p}(r) \le 1 \qquad \text{for every } r \in [1, \infty],$$

where $r = \infty$ means the point-input condition $\alpha q^{1-p} \le 1$. Moreover, every positive dual extremiser has one large coordinate, located at an atom of weight q, and all its other coordinates equal.

In plain words. To decide whether a number c bounds the coefficient, one only has to check a function of one real variable.

Theorem 2.5 (Closed formula and sharp threshold)**.** *Define*

$$\alpha_*(q) = q^{q/(1-q)}, \qquad c_0(q, \alpha) = \frac{\log(1/q)}{\log(1/(\alpha q))}. \tag{2.7}$$

(i) If $0 < \alpha \le \alpha_*(q)$, then

$$\eta(\rho, \alpha) = c_0(q, \alpha). \tag{2.8}$$

Except when $\rho = (1/2, 1/2)$ and $\alpha = 1/2$, the extremisers are exactly the point inputs δ_i with $\rho_i = q$.

(ii) If $\alpha > \alpha_*(q)$, then $\eta(\rho, \alpha) > c_0(q, \alpha)$, so no point input is an extremiser.

In plain words. Below the threshold the worst input simply announces the rarest symbol with certainty. Above it, it pays to hedge with a little erasure mass: the $\epsilon \log \epsilon$ gain in output entropy beats the cost.

For the uniform law on n symbols the threshold is $n^{-1/(n-1)}$ and (2.8) reads $\log n / \log(n/\alpha)$.

Proposition 2.6 (Local versus global)**.** *The local quadratic contraction coefficient of R_ρ at $m_{\rho,\alpha}$ is α, and $\eta(\rho, \alpha) < 1$. Moreover $\eta(\rho, \alpha) > \alpha$ except when $\rho = (1/2, 1/2)$ and $\alpha \ge 1/2$. (For the uniform binary prior, $\eta > \alpha$ still holds when $\alpha < 1/2$, by (A.1).)*

In plain words. Small perturbations of the reference shrink by the factor α, but large perturbations can shrink strictly less. For fixed α and $q \to 0$ we are below the threshold and $\eta = c_0 \to 1$, so the ratio of global to local coefficient approaches $1/\alpha$.

Theorem 2.7 (Extremisers above the threshold)**.** *Suppose $\rho \ne (1/2, 1/2)$ and $\alpha > \alpha_*(q)$. Then $\eta(\rho, \alpha) > \max\{\alpha, c_0(q, \alpha)\}$ and the supremum in (1.1) is attained. Every extremiser has full support. With $p_* = 1/\eta(\rho, \alpha)$, every extremiser is the following exponential tilt, for some index i with $\rho_i = q$ and some finite $r > 1$ satisfying*

$$H_{q,\alpha,p_*}(r) = 1, \qquad \partial_r H_{q,\alpha,p_*}(r) = 0 : \tag{2.9}$$

with $Z = \alpha(q r^{p_*} + 1 - q) + (1 - \alpha) r^{p_* q}$,

$$v_i = \frac{\alpha q r^{p_*}}{Z}, \quad v_j = \frac{\alpha \rho_j}{Z} \ (j \ne i), \quad v_e = \frac{(1 - \alpha) r^{p_* q}}{Z}. \tag{2.10}$$

Conversely, for every positive dual extremiser the tilt (2.10) is an extremiser.

In plain words. Above the threshold the worst input puts extra weight on one rarest symbol, keeps the other symbols in proportion, and keeps some erasure mass. No uniqueness of the scalar solution r is asserted, and a numerical solution of (2.9) certifies optimality only if it also passes the global test of Proposition 2.4.

3 Duality and the reduction to the smallest atom

3.1 Entropy duality

Lemma 3.1 (Finite variational duality)**.** *Suppose π and $\lambda = \pi K$ have full support and $c > 0$. Then $D(vK\|\lambda) \le cD(v\|\pi)$ for every input law v if and only if*

$$c \log \mathbb{E}_\pi \exp(Kf/c) \le \log \mathbb{E}_\lambda e^f \tag{3.1}$$

for every real output function f.

Proof. We use two forms of the Gibbs variational principle on a finite set. For a full-support law λ and any law u,

$$D(u\|\lambda) = \sup_f \{u(f) - \log \mathbb{E}_\lambda e^f\}, \tag{3.2}$$

where $u(f) = \sum_y u_y f_y$; the supremum is attained at $f = \log(u/\lambda)$ when u has full support, and is approached by finite f otherwise. For any real g,

$$\log \mathbb{E}_\pi e^g = \max_v \{v(g) - D(v\|\pi)\}, \tag{3.3}$$

attained exactly at $v \propto \pi e^g$.

Entropy inequality implies (3.1). Fix f. For every v, (3.2) with $u = vK$ gives $(vK)(f) - \log \mathbb{E}_\lambda e^f \le D(vK\|\lambda) \le cD(v\|\pi)$. Since $(vK)(f) = v(Kf)$, dividing by c gives $v(Kf/c) - D(v\|\pi) \le c^{-1} \log \mathbb{E}_\lambda e^f$ for every v. Taking the maximum over v and using (3.3) with $g = Kf/c$ gives (3.1).

(3.1) *implies the entropy inequality.* Fix v. For every f, (3.3) with $g = Kf/c$ and then (3.1) give

$$v(Kf) - cD(v\|\pi) \le c \log \mathbb{E}_\pi e^{Kf/c} \le \log \mathbb{E}_\lambda e^f.$$

Hence $(vK)(f) - \log \mathbb{E}_\lambda e^f \le cD(v\|\pi)$ for every f. Taking the supremum over f and using (3.2) with $u = vK$ gives $D(vK\|\lambda) \le cD(v\|\pi)$. □

For reverse erasure, $(R_\rho f)(i) = f_i$ and $(R_\rho f)(e) = \mathbb{E}_\rho f$. With $p = 1/c$ and $z_i = e^{f_i}$, $\mathbb{E}_{m_{\rho,\alpha}} \exp(pR_\rho f) = \alpha \sum_i \rho_i z_i^p + (1-\alpha) G_\rho(z)^p$, so (3.1) is exactly (2.4) for positive z; the extension to $z_i \ge 0$ follows by continuity because every $\rho_i > 0$. If (2.4) is strict for every positive nonconstant z, then the entropy inequality is strict at every input whose output has full support and differs from ρ: use the attaining function $f = \log((vR_\rho)/\rho)$ in the second half of the proof.

3.2 Shape of a maximising vector

Lemma 3.2 (One large coordinate)**.** *Let $p > 1$, $B > 0$, and maximise*

$$F(z) = \sum_i \rho_i z_i^p + BG_\rho(z)^p \quad \text{subject to } z_i \ge 0, \quad \sum_i \rho_i z_i = 1.$$

Every global maximiser lying on the boundary of this simplex is a vertex. Every nonconstant interior local maximiser has one large coordinate and all other coordinates equal.

Proof. The feasible set is compact. On its boundary some $z_i = 0$, so the product term vanishes. On a face of positive dimension the strictly convex power sum is maximised only at vertices of that face, which are vertices of the simplex. At an interior stationary point, Lagrange differentiation gives, with multiplier κ,

$$z_i^{p-1} + T/z_i = \kappa, \qquad T = BG_\rho(z)^p > 0$$

(after dividing by $p\rho_i$). Equivalently $x = z_i$ solves $x^p - \kappa x + T = 0$. This strictly convex function of $x > 0$ has at most two positive roots. If these are $x < y$, then $\kappa = (y^p - x^p)/(y - x)$ and $T = \kappa y - y^p$, so

$$T = xy \frac{y^{p-1} - x^{p-1}}{y - x} < (p-1)y^p,$$

because the chord slope κ is strictly below py^{p-1}. If at least two coordinates take the value y, choose nonzero h supported on them with $\sum_i \rho_i h_i = 0$. The second derivative of $F(z+sh)$ at $s=0$ is

$$\frac{p}{y^2}((p-1)y^p - T)\sum_i \rho_i h_i^2 > 0.$$

This is incompatible with a local maximum. Thus the larger root occurs at one coordinate only. □

The same reduction to a one-large-coordinate family, in primal form for the uniform Potts channel, is due to Gu and Polyanskiy [10, Prop. 17]; CCGP use it in their proof for $n = 3$. Lemma 3.2 is its dual, log-Laplace counterpart with arbitrary weights.

3.3 Which coordinate can be large?

Lemma 3.3 (Comparison at fixed arithmetic and geometric means)**.** *Fix $g \in (0,1)$ and $p > 1$. For $0 < w < 1$ let $a < b$ be the unique positive values satisfying*

$$(1-w)a + wb = 1, \qquad a^{1-w}b^w = g.$$

Then $\Psi_p(w) = (1-w)a^p + wb^p$ is strictly decreasing in w.

Proof. Existence and smoothness. Put $r = b/a > 1$. The equations give

$$a = \frac{1}{1-w+wr}, \qquad g = \frac{r^w}{1-w+wr}.$$

For fixed w, $\frac{d}{dr}\log\frac{r^w}{1-w+wr} = \frac{w(1-w)(1-r)}{r(1-w+wr)} < 0$ for $r > 1$, and the expression decreases from 1 to 0. Hence $r = r(w)$, and so a, b, exist uniquely and depend smoothly on w.

The derivative. Write primes for d/dw. Differentiating the two constraints gives

$$(1-w)a' + wb' = a - b, \qquad (1-w)\frac{a'}{a} + w\frac{b'}{b} = \log\frac{a}{b} = -\log r.$$

Put $A = (1-w)a'$ and $B = wb'$. Then $A + B = a(1-r)$ and $A + B/r = -a\log r$. Subtracting, $B(1-1/r) = a(1-r+\log r)$, so

$$B = \frac{ar(1-r+\log r)}{r-1}, \qquad A = a(1-r) - B = \frac{a\big((r-1) - r\log r\big)}{r-1}.$$

Now $\Psi_p' = b^p - a^p + pa^{p-1}A + pb^{p-1}B$. Substituting $b = ar$,

$$\begin{aligned}\Psi_p'(w) &= \frac{a^p}{r-1}\Big[(r^p-1)(r-1) + p(r-1) - pr\log r + pr^p(1-r+\log r)\Big] \\ &= \frac{a^p}{r-1}\Big[(r-1)\big(r^p - 1 - p(r^p-1)\big) + p\log r\,(r^p - r)\Big],\end{aligned}$$

that is,

$$\Psi_p'(w) = \frac{a^p}{r-1}\left[p\log r\,(r^p - r) - (p-1)(r-1)(r^p-1)\right]. \tag{3.4}$$

The sign. Set $I_1 = \int_1^r t^{p-1}\,dt = (r^p-1)/p$ and $I_0 = \int_1^r t^{p-2}\,dt = (r^{p-1}-1)/(p-1)$. Then $r^p - r = r(p-1)I_0$ and $r^p - 1 = pI_1$, so the bracket equals $p(p-1)\big[r\log r\, I_0 - (r-1)I_1\big]$. It is negative if and only if

$$\frac{I_1}{I_0} > \frac{r\log r}{r-1}.$$

Under the probability measure proportional to dt/t on $[1, r]$, the functions t and t^{p-1} are strictly increasing, so they have positive covariance. Their means are $L := (r-1)/\log r$ and $I_0/\log r$, and the mean of their product is $I_1/\log r$. Hence $I_1/I_0 > L$. The logarithmic mean satisfies $L > \sqrt{r}$: writing $r = e^{2s}$, this is $\sinh s > s$ for $s > 0$. Hence $L > r/L = r\log r/(r-1)$, which is the required strict sign. □

Proof of Theorem 2.3 and Proposition 2.4. Fix $p > 1$. By homogeneity, (2.4) holds if and only if $F(z) \le 1/\alpha$ on the simplex $\{z \ge 0 : \sum_i \rho_i z_i = 1\}$, where F is as in Lemma 3.2 with $B = (1-\alpha)/\alpha$. The maximum exists by compactness, and is attained at a vertex, at the constant vector, or at a nonconstant interior point.

Vertices. The vertex at atom i is $z = \delta_i/\rho_i$, with $\alpha F = \alpha\rho_i^{1-p}$. The largest such value is $\alpha q^{1-p} = H_{q,\alpha,p}(\infty)$.

The constant vector gives $\alpha F = 1$, i.e. equality.

Nonconstant interior maxima. By Lemma 3.2, such a maximiser has one high coordinate, at an atom of weight w, and a common lower value elsewhere; its weighted arithmetic mean is 1 and its weighted geometric mean is some $g \in (0,1)$. Suppose $w > q$. Choose an atom of weight q and let z' be the two-valued vector whose high value sits at that atom, with the two values chosen (as in Lemma 3.3, with weight q in place of w) so that its weighted arithmetic mean is still 1 and its weighted geometric mean is still g. Then z' is feasible, has the same geometric term $BG_\rho^p = Bg^p$, and its power sum is $\Psi_p(q) > \Psi_p(w)$ by Lemma 3.3. So a maximiser cannot have its high coordinate at a heavier atom. With the high coordinate at weight q, homogeneity lets us write the vector as a multiple of $(r, 1, \dots, 1)$ with $r > 1$, and αF equals $H_{q,\alpha,p}(r)$.

Combining the three cases, (2.4) holds if and only if $H_{q,\alpha,p}(r) \le 1$ for all $r \in [1,\infty]$. With $p = 1/c$ and Lemma 3.1, this is Proposition 2.4's equivalence. The criterion involves ρ only through q. Applying the same argument to the binary prior $(q, 1-q)$, whose smallest atom is also q, gives the identical criterion. Hence the sets $\{c : \eta(\rho,\alpha) \le c\}$ and $\{c : \eta((q,1-q),\alpha) \le c\}$ coincide, which proves Theorem 2.3.

Finally, a positive dual extremiser is, after normalisation, an interior global maximiser of F at $p = p_*$, so the strict comparison above places its high coordinate at an atom of weight q, with all other coordinates equal. □

4 Gao's mean inequality and the sharp threshold

Lemma 4.1 (Gao's mean inequality, with equality)**.** *Let $q = \min_i \rho_i$, $p \ge 1/(1-q)$, and $z_i \ge 0$. Then*

$$\left(\sum_i \rho_i z_i\right)^p \ge q^{p-1} \sum_i \rho_i z_i^p + (1 - q^{p-1}) G_\rho(z)^p. \tag{4.1}$$

Except for $\rho = (1/2, 1/2)$, $p = 2$, equality holds exactly when all coordinates coincide or when the only positive coordinate has weight q. In the exception, the inequality is an identity.

Attribution. For $1/(1-q) \le p \le 2$, (4.1) is Gao [8, Theorem 1.2], with mean order and outer exponent both equal to p. It is equation (1.7) in the published article and (1.4) in the inspected preprint. For $p \ge 2$, it is Gao [7, Theorem 3.1], with $t = 1$, $r = p$ and variables replaced by z_i^p; see also its rearrangement in equation (3.7). These papers continue a line of weighted arithmetic–geometric mean inequalities going back to Diananda [6]; further parameter ranges are developed in Gao [9], which does not contain the reduction of Theorem 2.3. The proof below is included to make the equality analysis self-contained; it does not claim novelty for the inequality or its method.

Proof. By Lemmas 3.2 and 3.3, applied with $\alpha = q^{p-1}$, it suffices to check a vertex, the constant vector, and a vector with one high coordinate at weight q. The first two give equality. Put $N = 1/q \ge 2$, which need not be an integer. The remaining inequality, after division by q^p, is

$$F(r) := (r + N - 1)^p - r^p - (N-1) - (N^p - N) r^{p/N} \ge 0, \quad r \ge 1. \tag{4.2}$$

We have $F(1) = 0$. The ratio

$$\frac{(r+N-1)^{p-1} - r^{p-1}}{N^{p-1} - 1}$$

is the average of $\big((r+s)/(1+s)\big)^{p-2}$ over $0 \le s \le N-1$, with positive weights proportional to $(1+s)^{p-2}\,ds$. If $1 < p \le 2$, this average is at least r^{p-2}, because $(r+s)/(1+s) \le r$. The hypothesis $p \ge N/(N-1)$ implies $r^{p-2} \ge r^{p/N-1}$. If $p \ge 2$, the weighted arithmetic–geometric mean inequality gives

$$\frac{r+s}{1+s} \ge r^{1/(1+s)} \ge r^{1/N}.$$

The average is therefore at least $r^{(p-2)/N}$, which is at least $r^{p/N-1}$ because $N \ge 2$. In both cases

$$\frac{F'(r)}{p} = (r+N-1)^{p-1} - r^{p-1} - (N^{p-1}-1)r^{p/N-1} \ge 0.$$

For $r > 1$ the derivative is strictly positive except when $N = p = 2$: use the strict comparison $(r+s)/(1+s) < r$ for $s > 0$ if $p < 2$, the strict exponent comparison if $p = 2 < N$, and the strict arithmetic–geometric mean inequality if $p > 2$. Thus every positive nonconstant equality vector is excluded, apart from that exception.

For a boundary vector, put $t_i = \rho_i z_i$. Then

$$\left(\sum_i t_i\right)^p \ge \sum_i t_i^p \ge q^{p-1} \sum_i \rho_i^{1-p} t_i^p.$$

Equality for a nonzero vector requires exactly one positive t_i, with $\rho_i = q$. At $q = 1/2, p = 2$ both weights equal $1/2$, and the asserted identity is the binomial formula. □

Proof of Theorem 2.5. Lower bound. A point input at a smallest atom, δ_i with $\rho_i = q$, has output δ_i, output divergence $\log(1/q)$ and input divergence $\log(1/(\alpha q))$. This gives $\eta \ge c_0$.

Upper bound for $\alpha \le \alpha_*(q)$. Set

$$p = 1/c_0 = \frac{\log(1/(\alpha q))}{\log(1/q)}, \quad \text{so that} \quad \alpha = q^{p-1}.$$

The condition $\alpha \le \alpha_*(q)$ is equivalent to $p \ge 1/(1-q)$. Then Lemma 4.1 is exactly the dual criterion (2.4), and Lemma 3.1 gives $\eta \le c_0$.

Equality cases. Exclude $\rho = (1/2, 1/2), \alpha = 1/2$. Let $v \ne m_{\rho,\alpha}$ with ordinary mass t. If $t < 1$, the output $vR_\rho = tu + (1-t)\rho$ has full support. If it differs from ρ, strictness in Lemma 4.1 and the remark after Lemma 3.1 make the entropy inequality strict; if it equals ρ, the ratio is zero. If $t = 1$, then $v = (u, 0)$ and the ratio is

$$\frac{D(u\|\rho)}{\log(1/\alpha) + D(u\|\rho)},$$

which increases strictly with $D(u\|\rho)$. Since $D(u\|\rho) = \sum_i u_i \log(1/\rho_i) - H(u) \le \log(1/q)$, with equality only for a point mass at an atom of weight q, the extremisers are exactly those point inputs.

Failure above the threshold. Take $v_\epsilon = (1-\epsilon)\delta_i + \epsilon\delta_e$ with $\rho_i = q$. As $\epsilon \downarrow 0$,

$$\begin{aligned} D(v_\epsilon R_\rho\|\rho) &= \log(1/q) + (1-q)\epsilon\log\epsilon + O(\epsilon), \\ D(v_\epsilon\|m_{\rho,\alpha}) &= \log(1/(\alpha q)) + \epsilon\log\epsilon + O(\epsilon). \end{aligned}$$

Hence the output divergence minus c_0 times the input divergence is

$$(1 - q - c_0)\epsilon\log\epsilon + O(\epsilon). \tag{4.3}$$

For $\alpha > \alpha_*(q)$ we have $c_0 > 1 - q$, so (4.3) is positive for all sufficiently small $\epsilon > 0$, and $\eta > c_0$. The point input at a smallest atom has the largest ratio among point inputs, so no point input is optimal. □

5 Local behaviour and the regime above the threshold

Proof of Proposition 2.6. Local coefficient. Write an input as $v = m_{\rho,\alpha} + h$ with $\sum_x h_x = 0$. On ordinary symbols write $h_i = \alpha\rho_i(b + g_i)$ with $\mathbb{E}_\rho g = 0$, and put $h_e = (1-\alpha)h'_e$. The constraint is $\alpha b + (1-\alpha)h'_e = 0$, and the output perturbation is $hR_\rho = (\alpha\rho_i g_i)_i$. Their χ^2 norms satisfy

$$\frac{\chi^2_\rho(hR_\rho)}{\chi^2_{m_{\rho,\alpha}}(h)} = \frac{\alpha^2\mathbb{E}_\rho g^2}{\alpha\mathbb{E}_\rho g^2 + \alpha b^2 + (1-\alpha){h'_e}^2} \le \alpha, \tag{5.1}$$

with equality when $b = h'_e = 0$. To pass from χ^2 to relative entropy uniformly, let $\|h\|$ be any norm and $\mu_0 = \min_x m_{\rho,\alpha}(x)$. From $(1+u)\log(1+u) = u + u^2/2 + O(|u|^3)$ for $|u| \le 1/2$,

$$D(v\|m_{\rho,\alpha}) = \tfrac{1}{2}\chi^2_{m_{\rho,\alpha}}(h) + O(\|h\|^3), \qquad D(vR_\rho\|\rho) = \tfrac{1}{2}\chi^2_\rho(hR_\rho) + O(\|h\|^3),$$

uniformly for all sufficiently small $\|h\|$ (so that every relative perturbation $h_x/m_{\rho,\alpha}(x)$ and $(hR_\rho)_i/\rho_i$ is at most $1/2$ in absolute value), with constants depending only on ρ and α. Also $\chi^2_{m_{\rho,\alpha}}(h) \ge C\|h\|^2$ for some $C > 0$. Hence the ratio is at most $(\alpha\chi^2_{m_{\rho,\alpha}}(h) + O(\|h\|^3))/(\chi^2_{m_{\rho,\alpha}}(h) - O(\|h\|^3)) = \alpha + O(\|h\|)$, and every sequence of inputs approaching $m_{\rho,\alpha}$ has ratio lim sup at most α. Choosing $b = h'_e = 0$ shows the value α is approached, so the local coefficient is α.

$\eta < 1$. Data processing is strict at every nonreference input. Indeed, equality in the data-processing inequality forces the posterior of the input given each occupied output to agree under v and under $m_{\rho,\alpha}$. If v gives erasure zero mass, its posterior differs from the reference posterior, which gives erasure positive mass. If v gives erasure positive mass, all outputs occur and posterior equality forces $v_i/(\alpha\rho_i) = v_e/(1-\alpha)$ for every i, hence $v = m_{\rho,\alpha}$. The ratio is continuous away from $m_{\rho,\alpha}$ on the compact simplex, is strictly below 1 there, and has lim sup $\le \alpha < 1$ at $m_{\rho,\alpha}$. Hence $\eta < 1$.

$\eta > \alpha$. Choose g with $\mathbb{E}_\rho g = 0$, $V = \mathbb{E}_\rho g^2 > 0$ and $m_3 = \mathbb{E}_\rho g^3 \ne 0$, and the input $v_s = (\alpha\rho_i(1+sg_i))_i \oplus (1-\alpha)$. Direct Taylor expansion gives

$$\frac{D(v_sR_\rho\|\rho)}{D(v_s\|m_{\rho,\alpha})} = \alpha\left(1 + (1-\alpha)\frac{m_3}{3V}s + O(s^2)\right). \tag{5.2}$$

Choosing the sign of s makes this greater than α. Such a g exists except for the uniform binary prior: take $g = \mathbf{1}_A - \rho(A)$ for an event of mass $w \notin \{0, 1/2, 1\}$; its third moment is $w(1-w)(1-2w)$. A singleton supplies such an event for every nonuniform binary law and for every law with at least three atoms. For the uniform binary prior, (A.1) gives $\eta > \alpha$ exactly when $\alpha < 1/2$. □

Proof of Theorem 2.7. Attainment and support. Proposition 2.6 and Theorem 2.5 give $\eta > \max\{\alpha, c_0\}$. By the local bound, a maximising sequence cannot approach $m_{\rho,\alpha}$, so compactness gives a nonreference maximiser v. Its erasure mass is positive: otherwise its ratio is at most c_0, as computed in the proof of Theorem 2.5. Its output is therefore positive. Put $f = \log((vR_\rho)/\rho)$, which attains (3.2) for $u = vR_\rho$. In the chain of inequalities in the second half of the proof of Lemma 3.1 (with $c = \eta$) both ends are equal, so every inequality is an equality. In particular (3.3) is attained, so $v \propto m_{\rho,\alpha}\exp(R_\rho f/\eta)$, a positive exponential tilt; hence v has full support. Also $z = e^f = (vR_\rho)/\rho$ gives equality in (3.1), i.e. it is a positive dual extremiser; it is nonconstant because $vR_\rho \ne \rho$ (the ratio is positive).

Form of the tilt. By Proposition 2.4 the dual extremiser is a multiple of $(r, 1, \dots, 1)$ with the large coordinate $r > 1$ at an atom i of weight q. The equality $H_{q,\alpha,p_*}(r) = 1$ is the equality in (2.4); since $H_{q,\alpha,p_*} \le 1$ on $[1,\infty]$ and r is an interior maximum, $\partial_r H_{q,\alpha,p_*}(r) = 0$. This is (2.9). Evaluating $v \propto m_{\rho,\alpha}\exp(p_*R_\rho f)$ with $e^f = (r, 1, \dots, 1)$ gives (2.10).

Converse. Let $z = e^f$ be a positive dual extremiser at p_*, and put

$$\Phi(f) = \log\mathbb{E}_\rho e^f - \eta\log\mathbb{E}_{m_{\rho,\alpha}} e^{R_\rho f/\eta}.$$

By Lemma 3.1, $\Phi \geq 0$ on $\mathbb{R}^n$, and $\Phi(f) = 0$. So f is an unconstrained global minimum and $\nabla\Phi(f) = 0$. Let $v \propto m_{\rho,\alpha} e^{R_\rho f/\eta}$ be the tilt. Since $\partial(R_\rho f)(x)/\partial f_i = R_\rho(x,i)$,

$$\frac{\partial\Phi}{\partial f_i} = \frac{\rho_i z_i}{\mathbb{E}_\rho z} - \sum_x v_x R_\rho(x,i) = \frac{\rho_i z_i}{\mathbb{E}_\rho z} - (vR_\rho)_i.$$

Hence $vR_\rho = (\rho_i z_i/\mathbb{E}_\rho z)_i$. Then (3.2) is attained at f for $u = vR_\rho$, and (3.3) is attained at v for $g = R_\rho f/\eta$, so

$$D(vR_\rho\|\rho) = (vR_\rho)(f) - \log \mathbb{E}_\rho e^f = v(R_\rho f) - \eta \log \mathbb{E}_{m_{\rho,\alpha}} e^{R_\rho f/\eta} = \eta D(v\|m_{\rho,\alpha}),$$

using $\Phi(f) = 0$ in the middle step. As z is nonconstant, $vR_\rho \neq \rho$, so $v \neq m_{\rho,\alpha}$ and v is an extremiser. This completes the description without assuming uniqueness of r. □

Remark 5.1 (An exact computational specification). By Proposition 2.4, $\eta(\rho,\alpha) \leq 1/p$ holds exactly when the scalar criterion holds at p. Since $\{c : \eta \leq c\}$ is the interval $[\eta,\infty)$, the set of exponents $p > 1$ at which the criterion holds is the interval $(1, p_*]$ with $p_* = 1/\eta$. Hence

$$\eta(\rho,\alpha) = \frac{1}{p_*}, \qquad p_* = \max\{p > 1 : H_{q,\alpha,p}(r) \leq 1 \text{ for all } r \in [1,\infty]\}.$$

This is a one-variable global optimisation with a scalar exponent search. Theorem 2.5 covers its whole point-optimal regime; a closed elementary formula or a uniqueness theorem in the remaining regime is not claimed.

Remark 5.2 (How small is $\eta - c_0$ just above the threshold?). Although Theorem 2.5(ii) is strict, the gain can be astronomically small. Expanding the boundary family v_ϵ one order further than (4.3) gives

$$D(v_\epsilon R_\rho\|\rho) - c_0 D(v_\epsilon\|m_{\rho,\alpha}) = \epsilon\big(\delta\log(1/\epsilon) + K\big) + O(\epsilon^2),$$

with $\delta = c_0 - (1-q)$ and $K = q\log(1/q) + \delta + c_0\log(1-\alpha)$. Just above the threshold δ is small and positive while $K < 0$ in all cases we computed, so the best member of this family has $\epsilon \approx e^{-1-|K|/\delta}$ and improves the ratio by roughly $\delta e^{-1-|K|/\delta}/\log(1/(\alpha q))$. Since δ is of order $\alpha - \alpha_*(q)$, this lower bound is exponentially small in $1/(\alpha - \alpha_*(q))$. Table 2 compares it with a numerical estimate of the excess computed from Remark 5.1 in 60-digit arithmetic. These are numerical observations, not theorems; in particular we do not prove an upper bound of the same exponential type. A practical consequence: a double-precision optimiser cannot see the change of regime for small q, and returns c_0 even where it is not the coefficient.

q	$\alpha_*(q)$	α	$\eta - c_0$ (numerical estimate)	best v_ϵ, $\epsilon = 10^{-k}$	heuristic
0.1	0.7743	0.85	7.9×10^{-22}	2.5×10^{-22}	3.1×10^{-22}
0.1	0.7743	0.90	5.9×10^{-16}	1.1×10^{-17}	1.4×10^{-17}
0.1	0.7743	0.95	2.3×10^{-13}	3.4×10^{-17}	3.4×10^{-17}
0.3	0.5969	0.65	3.3×10^{-7}	2.7×10^{-7}	2.8×10^{-7}
0.3	0.5969	0.70	3.2×10^{-5}	1.0×10^{-5}	1.6×10^{-5}
0.3	0.5969	0.80	5.0×10^{-4}	9.8×10^{-5}	9.8×10^{-5}
0.3	0.5969	0.90	1.0×10^{-3}	3.7×10^{-5}	5.6×10^{-5}

Table 2: Excess of the coefficient over the point-input value above the threshold (numerical; `verification/negative_controls.py`). The "heuristic" column is $\delta e^{-1-|K|/\delta}/\log(1/(\alpha q))$. Close to the threshold the boundary family captures the excess well; further away the true extremiser is a genuine tilt and the gap grows.

6 Bipartite reset chains and the conjecture

Let A, B be disjoint nonempty finite sets with positive conditional laws μ on A and ν on B. Consider the chain

$$P(a,a) = \tfrac{1}{2}, \quad P(a,b) = \tfrac{1}{2}\nu_b; \qquad P(b,b) = \tfrac{1}{2}, \quad P(b,a) = \tfrac{1}{2}\mu_a, \tag{6.1}$$

for $a \in A, b \in B$, with all other entries zero. Its stationary law is $\pi = (\mu/2, \nu/2)$, and detailed balance holds. Equivalently, this is the half-lazy walk on the complete bipartite graph with cross-edge weights $w_{ab} = \mu_a\nu_b$. For uniform μ, ν on parts of sizes m, n it is the half-lazy walk on $K_{m,n}$ (for $m \neq n$ the graph is not regular; π is then the degree-stationary law, with mass $1/2$ on each side). The cross weights have rank one; arbitrary weighted bipartite graphs are not covered.

For a part that is a singleton we set its minimum conditional mass to 1; so q below is effectively the minimum over the non-singleton parts.

Theorem 6.1 (Exact coefficient of half-lazy rank-one bipartite chains)**.** *If both parts are singletons, $\eta_{\mathrm{KL}}(\pi, P) = 0$. Otherwise, with $q = \min\{\min_a \mu_a, \min_b \nu_b\} \le 1/2$,*

$$\boxed{\eta_{\mathrm{KL}}(\pi, P) = \frac{\log(1/q)}{2\log(2/q)}.} \tag{6.2}$$

If $q < 1/2$, the only extremisers are the point inputs at vertices whose conditional probability is q. For uniform parts of sizes m, n, not both 1,

$$\eta_{\mathrm{KL}}(\pi, P) = \frac{\log M}{2\log(2M)}, \qquad M = \max(m, n). \tag{6.3}$$

In particular, for $m = n \ge 3$ the point inputs, and no other nonreference inputs, are extremisers. This proves the conjecture (1.2) together with its equality statement.

In plain words. The worst way to start the walk is at a single vertex of the side with the rarest vertex; the value depends only on that vertex's conditional probability. Starting anywhere else, or spreading the starting mass, loses strictly more entropy in one step.

Proof. Retention $1/2$ is always below the threshold. For $0 < q \le 1/2$, $\alpha_*(q) \ge 1/2$ is equivalent to $\phi(q) := (1-q)\log 2 + q\log q \ge 0$. Since $\phi'(q) = \log(eq/2) < 0$ on $(0, 1/2]$ and $\phi(1/2) = 0$, $\phi \ge 0$ there. So Theorem 2.5 applies at $\alpha = 1/2$ to every non-singleton conditional prior.

Part coefficients. Put

$$c_\mu = \frac{\log(1/q_\mu)}{\log(2/q_\mu)}, \quad q_\mu = \min_a \mu_a,$$

and similarly c_ν; set the coefficient of a singleton part to zero. Let $c = \log(1/q)/\log(2/q)$, so $c_\mu, c_\nu \le c$. By Theorem 2.5 and (2.3), $c_\mu = \eta(\mu, 1/2)$; for a singleton part the output is always μ and the bound below holds with coefficient zero.

A light-part bound. For $0 \le b \le 1/2$, convexity of $s \mapsto D(su + (1-s)\mu\|\mu)$ and the reverse-erasure inequality at input $(u/2, 1/2)$, which has ordinary mass $1/2$ and input divergence $\frac{1}{2}D(u\|\mu)$, give

$$D(bu + (1-b)\mu\|\mu) \le 2bD(\tfrac{u+\mu}{2}\|\mu) \le bc_\mu D(u\|\mu). \tag{6.4}$$

Decomposition. Write an input as $\xi = (tu_A, (1-t)u_B)$ with conditional laws u_A on A and u_B on B. Interchanging the parts if necessary, assume $t \ge 1/2$; this interchange swaps (μ, c_μ) with (ν, c_ν), which is harmless because only $c \ge \max\{c_\mu, c_\nu\}$ is used below. The input divergence is

$$D(\xi\|\pi) = d(t\|1/2) + tD(u_A\|\mu) + (1-t)D(u_B\|\nu). \tag{6.5}$$

The output assigns mass $1/2$ to each part, with conditional laws $tu_A + (1-t)\mu$ on A and $(1-t)u_B + t\nu$ on B. The first is the output of R_μ at the input $(tu_A, 1-t)$, whose divergence from $m_{\mu,1/2}$ is $U := d(t\|1/2) + tD(u_A\|\mu)$ by (2.3). Therefore, with $V := (1-t)D(u_B\|\nu)$,

$$\begin{aligned} 2D(\xi P\|\pi) &= D(tu_A + (1-t)\mu\|\mu) + D((1-t)u_B + t\nu\|\nu) \\ &\le c_\mu U + c_\nu V \le c\,(U+V) = c\,D(\xi\|\pi). \end{aligned}$$

The first inequality uses the reverse-erasure bound for the heavy part and (6.4) with $b = 1-t \le 1/2$ for the light part. The bookkeeping term $d(t\|1/2)$ is paid only once. A point input at a vertex of conditional mass q has input divergence $\log(2/q)$ and output divergence $\frac{1}{2}\log(1/q)$, so the bound $c/2$ is attained.

Equality when $q < 1/2$. Equality forces $(c-c_\mu)U = 0$, $(c-c_\nu)V = 0$ and equality in both part bounds. If $U > 0$, then $c_\mu = c$ (in particular the heavy part is not a singleton), and the heavy part attains its reverse-erasure coefficient at a nonreference input. Since $q_\mu = q < 1/2$, Theorem 2.5 forces $t = 1$ and u_A to be a point mass at an atom of weight q: a point input. (If the heavy part is a singleton, $c_\mu = 0 < c$, so $U = 0$ is forced; this case is covered next.) If $U = 0$, then $t = 1/2$ and $u_A = \mu$. A nonreference input then has $V > 0$, and equality requires $c_\nu = c$ and equality in (6.4) with $b = 1/2$, i.e. in the reverse-erasure bound at the input $(u_B/2, 1/2)$. That input has positive erasure mass and is not the reference, and $q_\nu = q < 1/2$, so the bound is strict by Theorem 2.5. This is impossible. The two-singleton chain resets to stationarity in one step, so its coefficient is zero. □

Corollary 6.2 (The lazy star)**.** *For the lazy random walk on the star with centre* 0 *and leaves* $1, \dots, N$, $N \ge 2$ *(stay with probability* $1/2$*; from the centre move to a uniform leaf; from a leaf move to the centre),*

$$\eta_{\mathrm{KL}}(\pi, P) = \frac{\log N}{2\log(2N)}.$$

For $N \ge 3$ the only extremisers are the point masses at leaves. For $N = 2$ (the path on three vertices) the extremisers are the family (6.7) *below.*

Proof. This is (6.1) with $A = \{0\}$, $\mu = \delta_0$ and ν uniform on the leaves, i.e. $K_{1,N}$; apply Theorem 6.1 and the paragraph on boundary equality below. □

In plain words. Jadhav [12, Proposition 9.5 and Remark 9.6] computed the point-mass value $\frac{1}{2}\log(n-1)/\log(2(n-1))$ for the lazy star on $n = N+1$ vertices and reported that numerical search up to $n = 50$ found nothing better, noting that this was evidence rather than proof. Corollary 6.2 proves it for every $n \ge 4$ and identifies the exceptional family at $n = 3$.

Boundary equality, spectrum and products

If $q = 1/2$, every non-singleton part consists of two equally likely states. The possibilities are $K_{2,2}$, $K_{1,2}$ and $K_{2,1}$, all with coefficient $1/4$. This is why CCGP restrict to $n \ge 3$: for $K_{2,2}$, all extremisers, up to exchanging the parts and relabelling within them, are

$$\big(s^2, (1-s)^2, s(1-s), s(1-s)\big), \qquad s \in [0,1] \setminus \{1/2\}. \tag{6.6}$$

For $K_{2,1}$, with the two-point part listed first, they are

$$\big(s^2, (1-s)^2, 2s(1-s)\big), \qquad s \in [0,1] \setminus \{1/2\}. \tag{6.7}$$

To see completeness, note first that a heavy singleton part forces $U = 0$ (as in the proof), hence $t = 1/2$; the light two-point part would then have to attain its reverse-erasure bound at an input with erasure mass $1/2$, and by the binary family of Appendix A this happens only at the reference. In the remaining cases the heavy part is the two-point part, and its reverse-erasure

equality family is the binary family proved in Appendix A. If $1/2 < t < 1$, strict convexity in (6.4) forces the light-part conditional law to be stationary. If $t = 1/2$, the binary reverse equality forces the heavy conditional law, and then the light conditional law, to be stationary; this is excluded. The endpoints $t = 1$ are the point inputs. These facts yield exactly (6.6)–(6.7). For $K_{1,1}$, every nonreference input has ratio zero.

The spectrum of (6.1) on $L^2(\pi)$ is 1 on constants, 0 on the part-difference function, and $1/2$ on functions with conditional mean zero within either part. When one part is nontrivial, the local quadratic coefficient is therefore $1/4$. In contrast, (6.2) tends to $1/2$ as $q \downarrow 0$: within this explicit family the global coefficient approaches twice the local one (Figure 1). For the reverse-erasure channel itself the ratio can be much larger, approaching $1/\alpha$ (Proposition 2.6); see [13, 17] for general comparison theory.

For synchronous products of these chains, the exact coefficient is the largest factor coefficient. This is standard KL tensorisation (Appendix B), not a new claim. Formula (6.2) concerns one step of laziness exactly $1/2$; Section 7 treats other laziness levels and powers.

7 Block-reset chains

The proof of Theorem 2.3 uses the reverse-erasure structure only through two facts: a test function enters through its arithmetic and geometric means, and a power sum is maximised at those means by a vector that singles out a rarest atom. This section shows that both facts survive in a much larger class of chains, with no restriction on the laziness. The consequence is that every chain in the class has the same coefficient as a small explicit chain, in which each block is replaced by a two-point law.

7.1 The class and its collapse

Let X be the disjoint union of finite nonempty blocks X_B, $B \in \mathcal{B}$, and let μ_B be a full-support law on X_B. Fix numbers $a_B \in [0,1]$ and $M_{BC} \geq 0$ with $\sum_C M_{BC} = 1 - a_B$. The *block-reset chain* is

$$K(x,\cdot) = a_B\,\delta_x + \sum_C M_{BC}\,\mu_C \qquad (x \in X_B). \tag{7.1}$$

From x it stays put with probability a_B; otherwise it chooses a block C and resamples a fresh point from μ_C. The cross weights between a point and a block are therefore rank one inside each pair of blocks. Take a reference law $\pi = \sum_B w_B \mu_B$ with $w_B > 0$, $\sum_B w_B = 1$. Stationarity is not required. Then

$$\lambda := \pi K = \sum_C w'_C\,\mu_C, \qquad w'_C = a_C w_C + \sum_B w_B M_{BC}. \tag{7.2}$$

We assume $w'_C > 0$ for every C, so that λ has full support; this holds automatically when $\pi K = \pi$. Write $q_B = \min \mu_B$. When $|X_B| \geq 2$, $q_B \leq 1/2$.

The *collapsed chain* $\widehat{K}$ has the same block indices, the same numbers a_B, M_{BC}, w_B, and block laws

$$\widehat{\mu}_B = (q_B,\, 1-q_B) \ \text{ on two points if } |X_B| \geq 2, \qquad \widehat{\mu}_B = (1) \ \text{ if } |X_B| = 1.$$

Its reference law is $\widehat{\pi} = \sum_B w_B \widehat{\mu}_B$. It has at most $2|\mathcal{B}|$ states, whatever the sizes of the original blocks.

Lemma 7.1 (Closure under composition)**.** *Let K and K' be block-reset chains on the same blocks and block laws, with coefficients (a, M) and (a', M'). Then KK' is a block-reset chain on the same blocks and laws, with*

$$a''_B = a_B a'_B, \qquad M''_{BD} = a_B M'_{BD} + M_{BD} a'_D + \sum_C M_{BC} M'_{CD}.$$

These coefficients do not involve the laws μ_B, so collapsing commutes with composition: $\widehat{KK'} = \widehat{K}\,\widehat{K'}$.

Proof. For $x \in X_B$, $\mu_C K' = a'_C \mu_C + \sum_D M'_{CD} \mu_D$. Substituting this into $KK'(x,\cdot) = a_B K'(x,\cdot) + \sum_C M_{BC}\, \mu_C K'$ gives the stated coefficients. □

In words. Running two block-reset steps in a row is again a block-reset step, so every power of a chain in the class is in the class.

7.2 Power sums at fixed arithmetic and geometric means

For $0 < w < 1$, $0 < g < 1$ and $p > 0$, let $a < b$ be the two positive values with $(1-w)a + wb = 1$ and $a^{1-w} b^w = g$, and put $\Psi_p(w) = (1-w)a^p + wb^p$, exactly as in Lemma 3.3. (Existence and uniqueness of $a < b$ is shown in the proof of that lemma and does not use p.) Here b is the larger value and w its weight.

Lemma 7.2 (Weight monotonicity for every exponent)**.** *Fix $0 < g < 1$. The function $w \mapsto \Psi_p(w)$ on $(0,1)$ is strictly decreasing if $p > 1$, strictly increasing if $0 < p < 1$, and constant equal to 1 if $p = 1$.*

Proof. The case $p = 1$ is the constraint itself. Let $p \neq 1$, $r = b/a > 1$, and write primes for derivatives in w at fixed g. Put $\xi_1 = (1-w)a'$ and $\xi_2 = wb'$ (called A and B in Lemma 3.3). Differentiating the two constraints gives

$$\xi_1 + \xi_2 = a - b, \qquad \frac{\xi_1}{a} + \frac{\xi_2}{b} = -\log r.$$

Solving this linear system,

$$\xi_2 = -\frac{b\,(r - 1 - \log r)}{r-1}, \qquad \xi_1 = \frac{a\,(r - 1 - r\log r)}{r-1}.$$

Hence

$$\Psi'_p(w) = b^p - a^p + p(a^{p-1}\xi_1 + b^{p-1}\xi_2) = \frac{a^p}{r-1}\Big[p \log r\,(r^p - r) - (p-1)(r-1)(r^p - 1)\Big],$$

which is (3.4); the elimination holds for every real p. Put $I_1 = \int_1^r t^{p-1} dt$ and $I_0 = \int_1^r t^{p-2} dt$, so that $r^p - 1 = pI_1$ and $r^p - r = (p-1) r I_0$. The bracket equals

$$p(p-1)\big[r \log r\, I_0 - (r-1) I_1\big].$$

Let σ be the measure dt/t on $[1, r]$ and, for real s, let $\ell(s) = \int t^{s+1} d\sigma / \int t^s d\sigma$, the mean of t under the law proportional to $t^s\, d\sigma$. Then $\log \ell$ has derivative $\mathbb{E}_{s+1} \log t - \mathbb{E}_s \log t$, and $s \mapsto \mathbb{E}_s \log t$ is strictly increasing because its derivative is the variance of $\log t$ under the tilted law. Hence ℓ is strictly increasing. Now $I_1/I_0 = \ell(p-1)$ and $\ell(-1) = \log r/(1 - 1/r) = r \log r/(r-1)$. As $p > 0$, $\ell(p-1) > \ell(-1)$, that is $(r-1) I_1 > r \log r\, I_0$. The square bracket is therefore negative, and the sign of Ψ'_p is the sign of $-p(p-1)$. □

In words. With both means held fixed, a power above one prefers the large value to sit on as little mass as possible, and a power below one prefers the small value to sit on as little mass as possible.

Lemma 7.3 (Maximal power sum at fixed means)**.** *Let μ be a full-support law on a finite set with at least two points, $q = \min \mu$, $p > 0$, $p \neq 1$ and $0 < g < 1$. Put*

$$S_p(\mu, g) = \sup\{\mathbb{E}_\mu y^p :\ y > 0,\ \mathbb{E}_\mu y = 1,\ G_\mu(y) = g\}, \qquad G_\mu(y) = \textstyle\prod_i y_i^{\mu_i}.$$

Then the supremum is attained and

$$S_p(\mu, g) = \begin{cases} \Psi_p(q), & p > 1, \\ \Psi_p(1-q), & 0 < p < 1. \end{cases}$$

Every maximiser takes exactly two values. For $p > 1$, the larger value occurs at exactly one coordinate; for $0 < p < 1$, the smaller value occurs at exactly one coordinate. In both cases that coordinate has mass q. In particular $S_p(\mu, g)$ depends on μ only through q, and $S_p(\mu, g) = S_p((q, 1-q), g)$. Trivially $S_1(\mu, g) = 1$, and at $g = 1$ the only feasible vector is $y \equiv 1$.

Proof. Compactness. On the feasible set F, $y_i \le 1/\mu_i$, and $y_j^{\mu_j} = g \prod_{i \ne j} y_i^{-\mu_i} \ge g \prod_{i \ne j} \mu_i^{\mu_i}$. So F is a closed subset of a compact box in $(0, \infty)^X$; the continuous objective attains its maximum.

Stationarity. The constraint gradients are (μ_i) and $(g\mu_i/y_i)$. They are linearly dependent only if y is constant, which forces $g = 1$. Hence F is a smooth submanifold of codimension two near every point, and a maximiser satisfies $p\mu_i y_i^{p-1} = \kappa_1 \mu_i + \kappa_2 g \mu_i / y_i$ for some multipliers κ_1, κ_2. With $C = \kappa_2 g$, every coordinate is a positive zero of

$$\phi(t) = pt^p - \kappa_1 t - C.$$

As y is not constant, ϕ has at least two positive zeros. If $p > 1$, ϕ is strictly convex with $\phi(0) = -C$; a strictly convex function with $\phi(0) \le 0$ has at most one positive zero, so $C < 0$. If $0 < p < 1$, ϕ is strictly concave and the same argument gives $C > 0$. A strictly convex or concave function has at most two zeros, so y takes exactly two values $x < z$.

One distinguished coordinate. Put $\psi(t) = \phi(t)/t$ and $Q(t) = p(p-1)t^p + C$, so that $\psi'(t) = Q(t)/t^2$. If $p > 1$, Q increases strictly from $C < 0$ to $+\infty$; if $p < 1$, it decreases strictly from $C > 0$ to $-\infty$. Either way Q has one zero t_0, ψ is strictly monotone on each side of t_0, and $\psi(x) = \psi(z) = 0$ forces $x < t_0 < z$. Consequently $Q(z) > 0$ if $p > 1$, and $Q(x) > 0$ if $p < 1$. Call that value v_* (z or x).

Suppose two coordinates $i \ne j$ both equal v_*. Take $h = e_i/\mu_i - e_j/\mu_j$. Then $\sum_k \mu_k h_k = 0$ and $\sum_k \mu_k h_k / y_k = 0$, so h is tangent to F. With the Lagrangian $L = \mathbb{E}_\mu y^p - \kappa_1 \mathbb{E}_\mu y - \kappa_2 G_\mu(y)$, and $h^\mathsf{T} \nabla^2 G_\mu h = G_\mu[(\sum \mu_k h_k/y_k)^2 - \sum \mu_k h_k^2/y_k^2]$,

$$h^\mathsf{T} \nabla^2 L\, h = \sum_k \mu_k h_k^2 \Big(p(p-1) y_k^{p-2} + \frac{C}{y_k^2} \Big) = \frac{Q(v_*)}{v_*^2} \sum_k \mu_k h_k^2 > 0.$$

This violates the second-order necessary condition for a local maximum on a smooth constraint manifold. (Here $|X| \ge 3$, since the other value also occurs, so the manifold has positive dimension.) Hence v_* occurs at exactly one coordinate i.

Which coordinate. If $p > 1$, the maximiser is the two-valued vector with its high value at i, and its value is $\Psi_p(\mu_i)$. If $p < 1$, the high value has weight $1 - \mu_i$ and the value is $\Psi_p(1 - \mu_i)$. By Lemma 7.2, both are at most the value at $\mu_i = q$, with equality only when $\mu_i = q$; the vector built on a rarest atom is feasible. For a two-point law the two-valued form is automatic and the same comparison applies. □

In words. Among positive vectors with prescribed arithmetic and geometric means, the largest power sum is always obtained by making one rarest coordinate different from the rest; which way it differs depends only on whether the power is above or below one.

7.3 The reduction

Theorem 7.4 (Block-reset reduction)**.** *Let K be a block-reset chain (7.1) with reference $\pi = \sum_B w_B \mu_B$ and $w'_C > 0$ for all C. For every $c > 0$,*

$$\eta_{\mathrm{KL}}(\pi, K) \le c \iff \eta_{\mathrm{KL}}(\widehat{\pi}, \widehat{K}) \le c.$$

Consequently

$$\eta_{\mathrm{KL}}(\pi, K) = \eta_{\mathrm{KL}}(\widehat{\pi}, \widehat{K}). \tag{7.3}$$

The coefficient depends on the block laws only through which blocks are singletons and through their minimum masses q_B. Moreover, lifting an input $\widehat{v}$ of $\widehat{K}$ by placing its rare-point mass of block B on one rarest atom of μ_B, and its other mass of block B proportionally to μ_B on the remaining atoms, preserves both divergences. Hence every extremiser of $\widehat{K}$ lifts to an extremiser of K.

Proof. Both π and λ have full support, and so do $\widehat{\pi}$ and $\widehat{\lambda}$, whose block weights are again w_B and w'_C. Lemma 3.1 therefore applies to both chains.

Let f be real on X, $y = e^f$, and let y_B be the restriction of y to X_B. Write $A_B = \mathbb{E}_{\mu_B} y_B$ and $G_B = G_{\mu_B}(y_B)$. Since $\mathbb{E}_{\mu_C} f = \log G_C$, for $x \in X_B$,

$$Kf(x) = a_B f(x) + \sum_C M_{BC} \log G_C.$$

With $p_B = a_B/c \geq 0$ and $\Gamma_B = \prod_C G_C^{M_{BC}/c} > 0$, the dual criterion (3.1) reads

$$\Big(\sum_B w_B \, \Gamma_B \, \mathbb{E}_{\mu_B}[y_B^{p_B}] \Big)^c \leq \sum_C w'_C \, A_C \qquad \text{for all } y > 0. \tag{7.4}$$

The right side and every Γ_B depend on y only through the numbers $(A_B, G_B)_B$. The set of y with prescribed numbers is a product over blocks. The left side is continuous and increasing in each $\mathbb{E}_{\mu_B}[y_B^{p_B}]$, because $w_B \Gamma_B > 0$. Hence (7.4) is equivalent to the same inequality with $\mathbb{E}_{\mu_B}[y_B^{p_B}]$ replaced by

$$\sup\{\mathbb{E}_{\mu_B} u^{p_B} : \; u > 0, \; \mathbb{E}_{\mu_B} u = A_B, \; G_{\mu_B}(u) = G_B\} = A_B^{p_B} \, S_{p_B}(\mu_B, G_B/A_B),$$

for every admissible family $(A_B, G_B)_B$. For a block of size one, $G_B = A_B$ and the supremum is $A_B^{p_B}$. For a block of size at least two, the admissible pairs are $0 < G_B \leq A_B$, and by Lemma 7.3 (with the trivial cases $p_B \in \{0, 1\}$ or $G_B = A_B$) the supremum depends only on q_B, p_B, A_B and G_B. The collapsed chain has the same weights, the same coefficients, the same admissible pairs, and block laws with the same minima. So the reduced inequality, and hence (7.4), holds for K exactly when it holds for $\widehat{K}$. This proves the equivalence for each $c > 0$, and (7.3) follows by taking the infimum over admissible c.

For the lift v of $\widehat{v}$, on the non-rarest atoms of block B the input v is proportional to μ_B, and so is the reference π. The output vK on the corresponding atoms of block C is $a_C v + (\sum_B M_{BC} v(X_B))\mu_C$, again proportional to μ_C, as is λ. Grouping these atoms therefore loses no divergence, and the grouped laws are exactly $\widehat{v}$, $\widehat{\pi}$, $\widehat{v}\widehat{K}$ and $\widehat{\lambda}$. Both divergences agree, so the ratios agree; with (7.3), extremisers lift. □

In words. For any chain that either stays put or jumps to a fresh sample from some block law, the worst-case one-step entropy contraction is the same as for a chain with at most two states per block, and only the rarest point of each block matters.

Remark 7.5 (No restriction on the laziness). No restriction on the laziness is needed: when some exponent $p_B = a_B/c$ is below one, Lemma 7.3 still applies, with a single *low* coordinate at a rarest atom. This matters: for P_s below ($s < 1/4$) one has $\eta_{\mathrm{KL}} \geq (1-2s)^2 > s$, so the coefficient exceeds every stay probability. Numerically, 20 of the 24 random chains tested by `verification/extension_checks.py` have a reported coefficient above their smallest positive stay probability, and the full and collapsed numerical estimates agree in all 24 to within 10^{-15}. The theorem does not assert that every extremiser of K is a lift, nor does it cover cross weights that are not of the rank-one block form (7.1); for such weights the minimum masses alone cannot determine the coefficient.

7.4 Consequences

Throughout, for laws μ on a part A and ν on a part B, and $s \in [0, 1]$, let P_s be the *s-lazy bipartite reset chain*: it stays put with probability s and otherwise samples the opposite part,

$$P_s(a, a) = s, \quad P_s(a, b) = (1-s)\nu_b; \qquad P_s(b, b) = s, \quad P_s(b, a) = (1-s)\mu_a.$$

Its stationary law is $\pi = (\mu/2, \nu/2)$, and $P_{1/2}$ is the chain (6.1). It is block-reset with $a_A = a_B = s$, $M_{AB} = M_{BA} = 1 - s$, $M_{AA} = M_{BB} = 0$.

Corollary 7.6 (Four-state reductions)**.** *For every $s \in [0,1]$ and every $k \geq 1$, the coefficient $\eta_{\mathrm{KL}}(\pi, P_s^k)$ equals that of the collapsed chain with the same coefficients on at most four states: four when both parts are non-singletons, three when exactly one is a singleton, and two when both are singletons. Each non-singleton part law μ is replaced by $(q_\mu, 1-q_\mu)$, while each singleton is retained. In the block form, P_s^k stays with probability s^k and otherwise resets to its own part or to the other part. For $s = 1/2$,*

$$P_{1/2}^k = \Pi + 2^{-k}(I - Q), \tag{7.5}$$

where Π resets to π and Q resets within the current part, and

$$\eta_{\mathrm{KL}}(\pi, P_{1/2}^k) \leq \Big(\frac{\log(1/q)}{2\log(2/q)}\Big)^k, \qquad q = \min\{q_\mu, q_\nu\}.$$

Proof. Lemma 7.1 and Theorem 7.4. For (7.5), write J for the cross reset, so $J^2 = Q$, $QJ = JQ = J$, and $\Pi = (J+Q)/2$. Then $P_{1/2} = (I+J)/2$ and induction gives $P_{1/2}^{k+1} = \frac{1}{2}(\Pi + 2^{-k}(I - Q))(I+J) = \Pi + 2^{-k-1}(I-Q)$, using $\Pi J = \Pi$ and $(I-Q)J = 0$. The bound follows from the chain rule $\eta_{\mathrm{KL}}(\pi, K_1K_2) \leq \eta_{\mathrm{KL}}(\pi, K_1)\,\eta_{\mathrm{KL}}(\pi K_1, K_2)$, stationarity of π, and Theorem 6.1. □

In words. Any laziness, and any number of steps, on a two-part reset chain is captured exactly by a chain with at most four states.

Remark 7.7 (Consistency with Theorem 6.1). At $s = 1/2$, Corollary 7.6 says that Theorem 6.1 for arbitrary part laws follows from its case of two-point parts. It is a consistency check rather than a new proof: the four-state value still has to be computed, and Theorem 6.1 says more, namely that only $\min\{q_\mu, q_\nu\}$ matters at this laziness. Numerically, full chains with parts $(0.1, 0.2, 0.7)$, $(0.3, 0.7)$, $(0.2, 0.3, 0.5)$, $(0.15, 0.25, 0.6)$ and uniform $K_{3,3}$ reproduce (6.2) to 10^{-15}.

Remark 7.8 (The lazy star at any laziness). For the star with $n \geq 3$ vertices (one singleton part, the centre, and leaves with uniform law) and any laziness $s \in [0,1]$, Corollary 7.6 shows that the coefficient equals that of the three-state chain with leaf law $(1/(n-1),\, 1 - 1/(n-1))$. For $s = 1/2$ see Corollary 6.2, which gives the value and the extremisers.

Corollary 7.9 (Complete multipartite graphs)**.** *Let G be the complete multipartite graph with $r \geq 2$ parts of sizes $n_1, \ldots, n_r$, $N = \sum_i n_i$, and let P stay put with probability s and otherwise move to a uniform neighbour. With its stationary law $\pi \propto$ degree, $\eta_{\mathrm{KL}}(\pi, P)$ equals the coefficient of the block-reset chain on at most $2r$ states with part laws $(1/n_i, 1 - 1/n_i)$ (a single state when $n_i = 1$), stay probability s, reset coefficients $M_{ij} = (1-s)n_j/(N - n_i)$ for $j \neq i$, and block weights $w_i \propto n_i(N - n_i)$.*

Proof. From a vertex of part i, the walk moves to part $j \neq i$ with probability $(1-s)n_j/(N-n_i)$ and then to a uniform vertex of that part; the degree of every vertex of part i is $N - n_i$. Apply Theorem 7.4. □

In words. A walk on a complete multipartite graph has the same one-step entropy contraction as a walk on two states per part.

7.5 Where point inputs stop being optimal

The following two facts are proved; the window they bound is then described numerically. For $s \in [0,1)$ and a vertex x of conditional mass ρ_x, the point ratio of P_s is

$$c_0(\rho_x, s) = \frac{s\log(2s/\rho_x) + (1-s)\log(2(1-s))}{\log(2/\rho_x)} = s - \frac{h(s) - (1-s)\log 2}{\log(2/\rho_x)},$$

with $h(s) = -s\log s - (1-s)\log(1-s)$. For $s \in [1/2, 1)$, $h(s) \geq 2(1-s)\log 2 > (1-s)\log 2$ by concavity of h, so the largest point ratio is at a vertex of the smallest conditional mass $q = \min\{q_\mu, q_\nu\}$.

Proposition 7.10 (Two obstructions to point optimality)**.** *(i) Assume the part containing a rarest vertex has at least two vertices. If $1/2 \le s < 1$ and $c_0(q,s) > (1-s)(1-q)$, then $\eta_{\mathrm{KL}}(\pi, P_s) > c_0(q,s)$; no point input is an extremiser.*

(ii) If $(1-2s)^2 > \max_x c_0(\rho_x, s)$, then $\eta_{\mathrm{KL}}(\pi, P_s) \ge (1-2s)^2$ exceeds every point ratio, so again no point input is an extremiser.

Proof. (i) Let a be a rarest vertex, in part A, and b any vertex of the other part. Put $v_\epsilon = (1-\epsilon)\delta_a + \epsilon\delta_b$. Its output gives mass $\epsilon(1-s)\mu_j$ to each $j \in A \setminus \{a\}$, which the point output does not charge; every other output coordinate is a smooth function of ϵ and positive at $\epsilon = 0$. Hence

$$D(v_\epsilon P_s \| \pi) = D(\delta_a P_s \| \pi) + (1-s)(1-q)\,\epsilon \log \epsilon + O(\epsilon), \qquad D(v_\epsilon \| \pi) = \log(2/q) + \epsilon \log \epsilon + O(\epsilon).$$

The output divergence minus $c_0(q,s)$ times the input divergence is $\big((1-s)(1-q) - c_0(q,s)\big)\epsilon \log \epsilon + O(\epsilon)$, which is positive for small $\epsilon > 0$ under the hypothesis. Since $c_0(q,s)$ is the largest point ratio, no point input is optimal. This is the argument of (4.3).

(ii) The chain is reversible with respect to π, and $\phi = \mathbf{1}_A - \mathbf{1}_B$ has $\mathbb{E}_\pi \phi = 0$ and $P_s\phi = (2s-1)\phi$. For $v_t = \pi(1 + t\phi)$, reversibility gives $v_t P_s = \pi(1 + t(2s-1)\phi)$. Expanding both divergences to second order, $D(v_t P_s \| \pi)/D(v_t \| \pi) \to (2s-1)^2$ as $t \to 0$. □

In words. Laziness a little above one half, or well below it, makes some spread-out starting law worse than every single vertex.

For uniform parts, $q = 1/n$, the root $s_+(n)$ of $c_0(q,s) = (1-s)(1-q)$ above $1/2$ is $1/2$ for $n = 2$ and 0.52054, 0.52885, 0.53307, 0.53877, 0.53228 for $n = 3, 4, 5, 10, 100$ (computed by root finding); the crossing $s_-(n)$ of c_0 with $(1-2s)^2$ is 0.26836, 0.26204, 0.25967, 0.25839, 0.25595, 0.25302 for $n = 2, 3, 4, 5, 10, 100$. The upper edge $s_+(n)$ is not monotone in n: it peaks near $n = 10$ and then decreases towards $1/2$. At $n = 3$ and 220-digit precision, the perturbation in (i) with $\epsilon = 10^{-200}$ gains 1.3×10^{-200} at $s = 0.525$ and loses 2.1×10^{-200} at $s = 0.515$, as (i) predicts.

Remark 7.11 (Numerical observations, not proved). We report the largest value among point-input evaluations, multistart numerical searches, and the local spectral lower bound. Accepted optimisation candidates are reevaluated at 40-digit working precision. Local spectral values may be limiting values rather than values attained at a nonreference input. These calculations provide numerical evidence, without certified error bounds or certified global optimality. They were produced by `verification/extension_checks.py` and cross-checked on the full chains.

(a) *Laziness window on $K_{n,n}$* ($n \in \{2, 3, 4, 5, 10, 100\}$, $s \in [0.20, 0.70]$ in steps of 0.01). For $s < 1/2$ the reported numerical estimate agrees with $\max\{c_0(1/n, s), (1-2s)^2\}$ to 10^{-15} at every grid point; no improvement over the point ratio larger than 10^{-9} is found at any grid point from the first one above $s_-(n)$ up to $1/2$. Above $1/2$, the first grid point with an improvement above 10^{-9} is $0.51, 0.53, 0.54, 0.55, 0.55, 0.55$ for $n = 2, 3, 4, 5, 10, 100$, and every later grid point also shows one. For $n = 2, 3$ this is the first grid point above $s_+(n)$, consistent with point optimality on exactly $[s_-(n), s_+(n)]$. For larger n it lies one or two grid steps above $s_+(n)$: just above $s_+(n)$ the gain from (i) is of $\epsilon \log \epsilon$ type with extremely small ϵ and is invisible in double precision, so there the grid detects failure only from outside and Proposition 7.10(i) is the sharper statement. For $n = 2$ the grid values above $1/2$ agree with s^2. Full six- and eight-state chains agree with the four-state values to 10^{-15}, for example 0.340916 versus point 0.340026 at $(n, s) = (3, 0.55)$ and 0.389576 versus 0.379126 at $(3, 0.6)$.

(b) *Powers at $s = 1/2$.* Point inputs are not optimal for $P_{1/2}^2$ and $P_{1/2}^3$: $\eta_{\mathrm{KL}} \approx 0.0721278$ (point 0.0644755) and 0.0185000 (point 0.0164340) for $n = 3$; 0.0886389 (point 0.0829304) and 0.0242136 (point 0.0227244) for $n = 5$. The best input found on four states for $n = 3, k = 2$ lifts to a full-support input of $K_{3,3}$. All values are below the bound in Corollary 7.6.

(c) *Complete multipartite graphs at $s = 1/2$.* Unlike the bipartite case, point inputs fail to be optimal in every tested example: $K_{2,3,4}$: 0.310885 versus point 0.308287; $K_{3,3,3}$: 0.286790

versus 0.276803; $K_{1,2,3}$: 0.288081 versus 0.285553; $K_{2,2,2,3}$: 0.289083 versus 0.276421. Full and collapsed chains agree to 10^{-15}.

An analytic description of the window $[s_-(n), s_+(n)]$, and closed forms for powers and for multipartite graphs, are left open. The reductions above turn each into a fixed small-dimensional problem.

8 An operational consequence: agreement distillation

Why this problem. In *agreement distillation* two parties holding correlated data want to agree on a common uniformly random key. Without communication, the probability of agreeing on k bits decays exponentially in k (Bogdanov and Mossel [3]); with one-way communication the question is how many bits must be sent per bit of key. Guruswami and Radhakrishnan [11] pinned this down for binary symmetric and binary erasure sources, and left more general sources open; Zhao [19, Chapter 3] showed that the optimal rate is governed by a fixed-source contraction coefficient (the hypercontractivity quantity s^*) and evaluated the reverse binary-erasure case [19, Corollary 3.1.3]. The reverse-erasure source arises when the speaker holds an erased copy of the listener's data. Our contribution here is the general-prior rate and a direct proof.

Model. Alice and Bob observe independent copies of a pair (E, X), where $X \sim \rho$ and $E = X$ with probability α, otherwise $E = e$. Alice sees the erased sequence E^N; Bob sees the original sequence X^N. The number N of samples is unrestricted. Both may use private randomness, independent of everything else; there is no shared randomness. Alice sends Bob one message S of fixed length ℓ bits and outputs a label $U \in [2^k]$ or a failure symbol; conditional on non-failure, U is exactly uniform. Bob outputs a guess $\hat{U}$. We require the failure probability and $\Pr(\hat{U} \neq U)$ to tend to zero as $k \to \infty$. (With variable-length messages of at most ℓ bits, replace ℓ by $\ell + 1$ below; the rate is unchanged.) No secrecy is required. Entropies and mutual informations in this section are in bits; this does not change contraction coefficients.

This model asks for vanishing error, which is what permits a Fano converse. Guruswami–Radhakrishnan and Zhao work with constant or exponentially small success probability, where the converse needs the hypercontractive argument of [19, Theorem 3.1.1]; we do not attempt that here.

Corollary 8.1 (Exact communication rate)**.** *If $0 < \alpha \leq \alpha_*(q)$, the optimal asymptotic number of communicated bits per output bit in this model is*

$$1 - \eta(\rho, \alpha) = \frac{\log(1/\alpha)}{\log(1/(\alpha q))}. \tag{8.1}$$

The rate is achievable with $(1 - \eta)k + O(\log k)$ bits and $O(k2^k \log k)$ source samples. The sample count is exponential in the key length; no computational or sample-efficiency claim is made.

In plain words. When the party with the damaged copy speaks, the rarer the rarest symbol, the less she needs to say. At the uniform binary prior this is Zhao's rate [19, Corollary 3.1.3] (with erasure probability $1 - \alpha$).

Proof. Converse. Let W be Alice's private randomness, so U is a function of (E^N, W) with W independent of (E^N, X^N). Then $U - E^N - X^N$ is a Markov chain, so for each value u, $P_{X^N|U=u} = P_{E^N|U=u} R_\rho^{\otimes N}$, while $P_{X^N} = \rho^{\otimes N} = m_{\rho,\alpha}^{\otimes N} R_\rho^{\otimes N}$. By tensorisation (Appendix B) the coefficient of $(m_{\rho,\alpha}^{\otimes N}, R_\rho^{\otimes N})$ is η, so

$$I(U; X^N) = \mathbb{E}_U D(P_{X^N|U} \| \rho^{\otimes N}) \leq \eta \, \mathbb{E}_U D(P_{E^N|U} \| m_{\rho,\alpha}^{\otimes N}) = \eta I(U; E^N) \leq \eta H(U),$$

for every N. Bob's guess is a function of (S, X^N, W') with his private randomness W' independent of (U, X^N, S). By Fano's inequality, with $P_e = \Pr(\hat{U} \neq U)$ and $2^k + 1$ possible values of U,

$$H(U) \leq I(U; S, X^N, W') + 1 + P_e \log_2(2^k + 1) \leq I(U; X^N) + H(S) + 1 + P_e(k + 1).$$

Since $H(S) \leq \ell$, we obtain $(1 - \eta)H(U) \leq \ell + 1 + P_e(k + 1)$. Let δ_k be the failure probability. Conditioning on the failure indicator, $H(U) \geq (1 - \delta_k)k$. As $\delta_k, P_e \to 0$, $\ell \geq (1 - \eta)k - o(k)$.

Achievability. Fix a smallest atom i, put $L = \log(1/(\alpha q))$, $J = 2^k$, and $m = \lceil k \log 2/L \rceil$. A table contains J independent blocks, each of m samples. In block j, let A_j be the event that Alice sees only the symbol i, and B_j the event that Bob sees only i. Then

$$A_j \subseteq B_j, \qquad p_A = (\alpha q)^m \in [\alpha q/J,\, 1/J], \qquad p_B = q^m.$$

The probability of exactly one A_j in a table is $Jp_A(1 - p_A)^{J-1}$, which is bounded below by a positive constant depending only on αq. Use $T = O(\log k)$ independent tables. Alice selects the first table with exactly one A_j and outputs its index j; if there is none, she fails. The failure probability can be made $O(k^{-2})$, and by exchangeability the index is exactly uniform conditional on non-failure.

Partition each table's indices into fixed buckets of size at most $b_k = \lfloor 1/(kp_B) \rfloor$. For all sufficiently large k we have $1 \leq b_k \leq J$ (indeed $b_k \to \infty$ because $p_B = q^m = J^{-c_0 + o(1)}$ with $c_0 < 1$) and $p_A \leq 1/2$. Alice sends the selected table number and the bucket containing her index, using a fixed number of bits. Bob outputs the unique index in that bucket for which B_j occurs, failing if it is not unique. Given the selected index, every other index in the selected table is conditioned on A_j^c, and

$$\Pr(B_j \mid A_j^c) = \frac{p_B - p_A}{1 - p_A} \leq 2p_B.$$

Conditioning on earlier rejected tables does not affect this table. A union bound over the bucket gives conditional decoding error at most $b_k \cdot 2p_B \leq 2/k$. The message length is

$$\lceil \log_2 T \rceil + \lceil \log_2 \lceil J/b_k \rceil \rceil = k + m \log_2 q + O(\log k) = (1 - c_0)k + O(\log k).$$

The number of samples is $TJm = O(k2^k \log k)$. Theorem 2.5 identifies $c_0 = \eta$, proving optimality. □

Remark 8.2 (The opposite direction, and what is open). If instead the full-view party (Bob, holding X^N) speaks to the party holding the erased copy, the relevant channel is the forward erasure channel, whose KL contraction coefficient equals α for every input law (the output divergence is exactly α times the input divergence). The converse above then gives at least $1 - \alpha$ bits per key bit; we do not prove achievability in that direction. Since $\eta(\rho, \alpha) \geq \alpha$, the converse bound for the erased speaker is never larger, and below the threshold her optimal rate can be far smaller: for $\alpha = 0.5, q = 0.1$ the rate (8.1) is 0.231, against the lower bound 0.5 in the opposite direction; for $\alpha = 0.7, q = 0.05$ it is 0.106 against 0.3. The two coincide at $\alpha = q = 1/2$. For uniform binary sources this opposite direction is treated by Guruswami and Radhakrishnan [11]. Two questions remain open. Above the threshold the converse $1 - \eta$ still holds, but the scheme gives only $1 - c_0 > 1 - \eta$; the optimal rate there is open. And we do not know whether the rate can be approached with polynomially many samples.

9 Discussion and open problems

The proofs above are analytic. The bundle also contains finite numerical diagnostics (Appendix C), including negative controls showing that the checks detect false versions of the main claims. They are evidence of implementation correctness, not substitutes for proof.

The theorem resolves a specific published finite-chain conjecture, settles the lazy-star observation of [12], and gives a broader structural fact about reverse erasure. It does not resolve entropy contraction for arbitrary graphs or arbitrary edge weights. Natural next problems:

(1) *Above the threshold.* Prove uniqueness of the tangency solution r in (2.9), at least for binary priors, and determine the asymptotics of $\eta - c_0$ as $\alpha \downarrow \alpha_*(q)$ (an upper bound matching the exponential lower bound of Remark 5.2) and of $\eta - \alpha$ as $\alpha \uparrow 1$.

(2) *Laziness and powers.* Section 7 extends the reduction to block-reset chains; the observations recorded there about where point inputs stop being optimal invite an analytic description of the full phase diagram.

(3) *Beyond rank one.* The formula cannot extend to arbitrary cross weights: with two states per part and cross-transition matrix $\left(\begin{smallmatrix} 1-\epsilon & \epsilon \\ \epsilon & 1-\epsilon \end{smallmatrix}\right)$, the half-lazy chain has eigenvalue $1-\epsilon$, so $\eta_{\mathrm{KL}} \geq (1-\epsilon)^2$, whereas (6.2) gives $1/4$. Bounds in terms of the distance from rank one would be natural.

(4) *Other constants.* Compute the half-step and continuous-time (MLSI) constants of [4] for the same family, to place it fully in their comparison.

(5) *Communication.* Determine the agreement rate above the threshold, and whether polynomially many samples suffice.

(6) *Reference-optimised coefficient.* Decide whether the reduction to the smallest atom also controls the coefficient maximised over reference laws [15].

10 Research and verification status

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This paper was prepared with substantial AI assistance. Its checks are internal and AI-assisted only:

- *No human peer review.* Several independent AI reviewers (editorial, methodology, domain, cross-disciplinary and adversarial) read the proofs line by line and reproduced the main formulas by direct optimisation and high-precision arithmetic. They found no critical or major correctness gap; their comments on exposition, literature and tooling are addressed in this revision. Section 7 was added in revision; its proofs and numerical observations were checked in a separate AI verification re-review and proof audit. This is not peer review.
- *No formal proof.* No proof assistant was used. The arguments are informal mathematical proofs.
- *Numerical diagnostics are not proofs.* The scripts in the bundle test the claims on finite samples, structured families and negative controls (Appendix C); they do not certify any unrestricted statement.
- *Priority.* The literature search was bounded (arXiv abstract queries, web indexes, OpenAlex citers of [4]; one citation index was rate-limited). Every reference was copied from its Crossref, DataCite or arXiv record, or for [19] from the thesis title page; the numbering of [4] was checked in both the arXiv v1 text and the authors' manuscript.

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A The known uniform binary case

For completeness, the full known coefficient is

$$\eta((1/2,1/2),\alpha) = \begin{cases} \log 2/\log(2/\alpha), & 0 < \alpha \leq 1/2, \\ \alpha, & 1/2 \leq \alpha < 1. \end{cases} \tag{A.1}$$

The first branch is Zhao's [19, Theorem 3.5.4]. The second follows from Nair and Wang [14] as follows. With Y uniform binary and E its erased observation (erasure probability $1-\alpha$), their Theorem 1 characterises hypercontractivity of (Y,E) with parameters (λ_1,λ_2) by the inequality $D(\nu_Y\|\mu_Y)/\lambda_1 + D(\nu_E\|\mu_E)/\lambda_2 \le D(\nu_{YE}\|\mu_{YE})$ for all ν, and their Theorem 3 determines the parameters on the curve $(\lambda_1-1)(\lambda_2-1)=\alpha$. Choosing $\nu_{YE}=\nu_E\,\mu_{Y|E}$ gives $D(\nu_{YE}\|\mu_{YE}) = D(\nu_E\|\mu_E)$ and $\nu_Y=\nu_E R$, so

$$D(\nu_E R\|\mu_Y) \le \frac{\lambda_1(\lambda_2-1)}{\lambda_2}\, D(\nu_E\|\mu_E).$$

With $\lambda_2 = 1+s$ and $\lambda_1 = 1+\alpha/s$ the factor is $(\alpha+s)/(1+s)$. For $\alpha \ge 1/2$ the hypercontractivity condition holds for every $s>0$, and letting $s\downarrow 0$ gives $\eta\le\alpha$; the local coefficient gives equality. We also include an elementary proof, which supplies the equality cases used above.

The first branch follows from Theorem 2.5. For $\alpha \ge 1/2$, put $p = 1/\alpha \in (1,2]$. By homogeneity, a positive binary vector can be written as (e^u, e^{-u}), $u\ge 0$. The dual inequality is

$$\cosh(pu) \le p\cosh(u)^p - p + 1.$$

Both sides equal one at $u=0$. Comparing derivatives reduces to $\sinh(pu) \le p\sinh u\cosh(u)^{p-1}$, or, with $y=\tanh u$,

$$(1+y)^p - (1-y)^p \le 2py.$$

The derivative of the left side is $p((1+y)^{p-1}+(1-y)^{p-1}) \le 2p$, by concavity. This proves the upper bound α, and the local coefficient gives the lower bound. For $1<p<2$, the positive nonconstant dual inequality is strict. Zero-erasure inputs have ratio at most $\log 2/\log(2/\alpha) < \alpha$ for $1/2<\alpha<1$, because the strictly concave function $\alpha\log(2/\alpha)$ equals $\log 2$ at both ends of $[1/2,1]$. Thus no nonreference input attains this branch's supremum.

At $\alpha = 1/2$, the dual identity is the binomial formula. If the output is $(s,1-s)$, equality in the input entropy variational formula gives exactly

$$v_s = (s^2, (1-s)^2, 2s(1-s)), \qquad s\in[0,1]\setminus\{1/2\}.$$

Direct substitution gives $v_sR = (s,1-s)$ and $D(v_s\|m) = 2D((s,1-s)\|(1/2,1/2))$. Positive outputs force this unique tilt, and boundary outputs give its endpoint point inputs. This proves completeness of the equality family.

B Standard KL tensorisation

For finite channels K_j with input references π_j, output references $\lambda_j = \pi_j K_j$, and coefficients η_j,

$$\eta_{\mathrm{KL}}\Big(\bigotimes_j \pi_j, \bigotimes_j K_j\Big) = \max_j \eta_j.$$

Here is a direct proof of the upper bound, also used in Section 8. Let arbitrary jointly distributed X_j pass through the channels independently: given X_j, the output Y_j is drawn from $K_j(X_j,\cdot)$ independently of everything else. Two conditional independences follow: Y_j is independent of $(X_{<j},Y_{<j})$ given X_j, so $\mathcal{L}(Y_j \mid Y_{<j}) = \mathcal{L}(X_j\mid Y_{<j})K_j$; and X_j is independent of $Y_{<j}$ given $X_{<j}$. Chain rules and the individual inequalities give

$$\begin{aligned} D(\mathcal{L}(Y)\|\textstyle\bigotimes_j \lambda_j) &= \sum_j \mathbb{E}\, D(\mathcal{L}(Y_j\mid Y_{<j})\|\lambda_j) \\ &\le \eta_{\max}\sum_j \mathbb{E}\, D(\mathcal{L}(X_j\mid Y_{<j})\|\pi_j) \\ &\le \eta_{\max}\sum_j \mathbb{E}\, D(\mathcal{L}(X_j\mid X_{<j})\|\pi_j) \\ &= \eta_{\max} D(\mathcal{L}(X)\|\textstyle\bigotimes_j \pi_j). \end{aligned}$$

The second inequality is convexity: by the second conditional independence, the law of X_j given $Y_{<j}$ is a mixture of its laws given $X_{<j}$. For the lower bound, vary one input coordinate only, keep the others at reference, and take a sequence approaching that coordinate's supremum. See also [16, 17].

C Numerical diagnostics

Four scripts in `verification/` produce the recorded outputs in `verification/results/`. Each collects failures explicitly (no `assert`), computes its status, exits with status 1 on any failure, writes only to a new output directory, and records its own SHA-256. Each was run twice from a scratch working directory, normally and with Python's `-O` flag, with identical outcomes.

Random-sample suite (`validate_entropy.py`, fixed seed). It evaluates the claims on full transition matrices and channels:

Diagnostic family	Inputs or tests
Complete-bipartite inputs, using full matrices	88,736
Uniform mean-inequality tests	126,000
Uniform reverse-erasure inputs below the threshold	108,218
Weighted reverse-erasure and reset-chain inputs	66,084
Binary endpoint identities	202
High-precision witnesses above the threshold	4
Exact rational kernel checks (stochasticity, stationarity, detailed balance)	5 kernels

The principal residual gate is 2×10^{-12}. Random samples rarely land near extremisers, so these counts mainly confirm the inequalities on generic inputs.

Negative controls and structured families (`negative_controls.py`). Ten false claims must be detected, and all are: Gao's inequality at $p = 0.97/(1-q)$ for three priors (detected in 60-digit arithmetic on the vectors $(r, 1, \dots, 1)$); the closed formula claimed slightly above and well above the threshold for four priors (detected by v_ϵ in 400-digit arithmetic, down to $\epsilon = 10^{-74}$); and a bipartite coefficient shrunk by 0.1% for three graphs (detected at point inputs). The random sampler alone would have detected only three of the ten. The structured families pass: v_ϵ never beats c_0 at or below the threshold; the large coordinate at the rarest atom dominates heavier atoms; the tilt (2.10) built from (2.9) reproduces the scalar-criterion value to more than 20 digits for five priors (two values of q), and identically for distinct priors with the same q, as Theorem 2.3 predicts; an unconstrained double-precision optimisation over the whole simplex agrees with the scalar criterion to about 10^{-15}; and Gao's weighted inequality holds on structured and random vectors. The near-threshold values in Table 2 come from this script.

Minimum-mass reduction (`verify_minimum_mass.py`). It compares sampled scalar envelopes with the large coordinate at the rarest atom and at a heavier atom (256 parameter choices), and power moments at fixed arithmetic and geometric means (896 comparisons). Of the 256 envelope comparisons, 111 are vacuous (both log-envelopes within 10^{-12} of zero, the constant-vector value), 2 are decided by the point-input limit, and 143 are informative. Over the informative cases the heavier-minus-rarest log-envelope difference is always negative (largest value -5.6×10^{-3}), and every moment comparison has the sign predicted by Lemma 3.3.

Block-reset extension (`extension_checks.py`). It records 122 checks and 5 negative controls (all detected): the power-sum lemma for exponents above and below one; 24 random block-reset chains, full against collapsed (agreement to 10^{-15}); the laziness grid on $K_{n,n}$; the powers $P_{1/2}^k$ and the closure identity; the four multipartite graphs; and the star at several laziness levels. Reported

coefficients are numerical estimates, as explained in Remark 7.11. It produces the numbers in Section 7.

A sensitivity example. For the uniform ten-symbol prior and $\alpha = 0.9$, the input $(1 - 10^{-15})\delta_1 + 10^{-15}\delta_e$ improves on the point ratio by about 1.1333×10^{-17}, positive at 120-digit precision. A floating-point optimiser misses it, which is why the threshold proof uses the $\epsilon \log \epsilon$ term (Remark 5.2).

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